MILITARY SPECIFICATION

TERMINAL, TELEGRAPH-TELEPHONE AN/TCC-29( ) AND UNITS OF

1. SCOPE

1.1 This specification covers Terminal, Telegraph-Telephone AN/TCC-29( ) which consists of one of each of the following units:

Terminal, Telegraph TH-22( )/TG
Filter Assembly, Electrical F-316( )U
Converter, Telegraph-Telephone Signal CV-425( )/U

This is a tactical, transistorized terminal that provides one duplex channel for the simultaneous transmission of telegraph or teletypewriter pulses and speech in the voice frequency (VF) range (See 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, (And Connectors)
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-552197	Terminal, Telegraph TH-22( )/TG
SC-DL-552198	Filter Assembly, Electrical F-316( )/U
SC-DL-552199	Converter, Telegraph-Telephone Signal CV-425( )/U

(Copies of specification, standards, and drawings, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

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3. REQUIREMENTS

3.1 First article samples. - The contractor shall furnish two first article samples for approval, if required by the invitation for bids and contract (See 4.3 and 6.2(c)).

3.2 Construction. - The equipment shall be constructed in accordance with the requirements of this specification and of the applicable drawing and data list/s as follows (including the requirements for parts, materials and processes thereon) (See 4.4 and 4.7):

Terminal, Telegraph TH-22( )/TG	SC-DL-552197
Filter Assembly, Electrical F-316( )/U	SC-DL-552198
Converter, Telegraph-Telephone Signal CV-425( )/U	SC-DL-552199

3.3 Parts, materials and processes; general. - Any parts, materials or processes to be substituted for those required by this specification (including subsidiary specifications and drawings herein) shall be selected in accordance with Specification MIL-P-11268 and submitted to the government for approval in accordance with the instructions therein.

3.4 Finish. - The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings (See 4.4).

3.5 Marking. -

3.5.1 General. - Marking shall conform to Specification MIL-M-13231 (See 4.4). Front panel marking shall be Group I as described in that specification.

3.5.2 Visibility. - Wherever practible, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. - All items listed in 1.1 shall have serial numbers.

3.6 Interchangeability. - Like units, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 (See 4.9).

3.7 Preconditioning. - The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1 followed by forty-eight hours of electrical preconditioning in accordance with 4.6.2. Prior to subjecting equipment to preconditioning, contractor shall have evidence of conformance to Group A inspection requirement.

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3.8 Electrical requirements for Terminal, Telegraph TH-22( )/TG. -

3.8.1 Power sources. - The terminal shall operate from power sources of 115 or 230 ± 10 percent volts ac, 50 to 60 ±3Hz and 26 ± 4 volts dc. (See 4.10.1)

3.8.1.1 Transients. - When operating from a 26 volt dc power source, the circuitry shall provide protection against voltage transients of 80 volts maximum with a time duration of 10 microseconds, and 40 volts with a time duration of 250 milliseconds occurring on the input power supply lines. (See 4.10.1.1)

3.8.1.2 Polarization: When operating from a 26 volt dc power source, the circuitry shall provide protection against damage to the equipment when the polarity of the power leads is reversed (See 4.10.1.2).

3.8.2 Vf output frequencies. - The terminal send circuit shall generate the following output frequencies:

Mark	1317.5 ± 3 Hz
Space	1232.5 ± 3 Hz

The average transmitted frequency of the terminal when pulsed by an input of mark and space impulses of equal length at the rate of 37.5 Hz shall be 1275 ± 3 Hz (See 4.10.2).

3.8.3 Vf output level. - The vf output level of the terminal into a 600 ohm resistive load shall be 0 dbm + 4 db - 2 db. The difference in level between mark and space frequencies shall not exceed 1 db (See 4.10.3).

3.8.4 Distortion. - The total peak distortion introduced into a teletypewriter signal being processed by the terminal shall not exceed five percent. This requirement applies at teletypewriter speeds of 60, 75, 100 and 200 words per minute (wpm) based on 7.42 unit intervals per operation and 6 operations per word. No internal adjustment of the terminal will be allowed when switching between modes (mode one provides for operation at 60, 75 and 100 wpm and mode two at 200 wpm)(See 4.10.4).

3.8.5 Signal delay. - Signal transitions shall be delayed by the terminal send circuit 8 ± 2 milliseconds in the MODE ONE switch position and 5 ± 1 milliseconds in the MODE TWO switch position (See 4.10.5)

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3.8.6 Carrier suppression. -

3.8.6.1 Normal signal. - With the NORM-REC-SEND switch in the NORM position, the marking tone shall be removed automatically during periods of no transmission. The carrier suppression time delay shall be such that a 1317.5 Hz signal shall persist for only 2.5 ± 0.5 seconds after the last space signal. After this time, the vf output shall be less than -55 dbm. Upon resumption of transmission, the circuit operation shall be such that marking tone is restored to the line before the initial mark to space transition of the first character is transmitted. The result shall be that no degradation of the first character or of the following characters shall occur. (See 4.10.6)

3.8.6.2 Send signal. - With the NORM-REC-SEND switch in the SEND position, there shall be no carrier suppression. The unit shall transmit a 1317.5 Hz signal continuously after the last space signal. (See 4.10.6)

3.8.6.3 Receive signal. - With the NORM-REC-SEND switch in the REC position, the vf output level shall be less than - 55 dbm (See 4.10.6).

3.8.7 Send circuit impedance. - The output impedance of the terminal measured across the 2W 4WR binding posts and across the 4WS TEL binding posts shall be 600 + 10% -20% ohms when transmitting a 1232.5 ± 3 Hz signal (See 4.10.7).

3.8.8 Send jack current. - The current measured at the SEND jack shall be not less than 16.5 milliamperes (ma) (See 4.10.8).

3.8.9 DC loop output, internal power. - The receive circuit shall supply 130 volt (nominal) dc power to external loads over a range of 0 to 1000 ohms through the REC jack. Over the range 100 to 1000 ohms, mark current shall be adjustable to 21 ± 2 ma; space current shall not exceed 1.0 ma. With a short circuit load, the loop current shall not exceed 26 ma, and no damage shall occur to the terminal. (See 4.10.9)

3.8.10 DC loop output, external power. - The receive circuit shall key a 130 volt dc, 60 ma, 1000 ohm externally powered loop connected to the terminal KEYING jack (See 4.10.10).

3.8.11 Receive circuit sensitivity. - The vf input level required to key the terminal receive circuit shall be 0 to -45 dbm with the high-low sensitivity strap in the high sensitivity position (See 4.10.11).

3.8.12 Receive circuit impedance. - The impedance measured across the 2W 4WR binding posts, with power applied to the terminal, shall be 600 +10% -20% ohms over the frequency range 1000 to 2000 Hz (See 4.10.12).

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3.8.13 Threshold circuit. - The receive circuit output shall switch to steady mark signal, and the THRESHOLD lamp shall light when the input signal level falls to -25 dbm ± 3 db with the high-low sensitivity strap attached in the low sensitivity position or -50 dbm ± 5 db with the strap attached in the high sensitivity position. The output shall reach steady mark condition not more than 2 milliseconds after the input level falls to the threshold level (to avoid activating teletypewriters connected to the receive circuit that are in motor stop condition) (See 4.10.13).

3.8.14 Break-in circuit. -

3.8.14.1 Break-in transmission. - With the 4W-2W-TEL switch in the 2W position, the NORM-REC-SEND switch in the NORM position and the terminal receiving a teletypewriter signal (carrier suppression circuit activated) in a two wire (half-duplex) circuit, operation of the BREAK IN switch shall cause transmission of an 1180 ± 3 Hz break-in signal. Wiring the break-in disable strap in place shall prevent operation of the break-in circuit. (See 4.10.14).

3.8.14.1.1 Break-in signal level. - The break-in signal level shall be 0dbm + 4 db - 2 db with the BOOST Switch in the lower position and 10 dbm + 4 db - 2 db with the BOOST Switch in the upper (operated) position (See 4.10.14).

3.8.14.1.2 Break-in circuit lock out. - Transmission of the break-in signal shall lock out the originating terminal's break-in circuit and place the terminal in stand by condition (steady mark signal on KEYING and REC jacks and carrier suppression circuit activated) (See 4.10.14).

3.8.14.2 Break-in interference. - The break-in signal shall not be audible in nor interfere with the telephone circuits of the terminals originating and receiving the signal (See 4.10.15).

3.8.14.3 Break-in reception. - With the 4W-2W-TEL switch in the 2W position, the NORM-REC-SEND switch in the NORM position and the terminal transmitting a teletypewriter signal into a two wire (half-duplex) circuit, reception of an 1180 ± 3 Hz break-in signal shall switch the terminal into ring condition (steady space signal on KEYING and REC jacks; steady space signal on vf output, carrier suppression circuit disabled and audible alarm activated) (See 4.10.16).

3.8.14.4 Break-in sensitivity. - The break-in circuit shall switch the terminal receiving the break-in signal into ring condition with input signal levels down to -31 dbm (See 4.10.16).

3.8.14.5 Break-in time delay. - An 1180 ± 3 Hz break-in signal of less than .25 second in duration shall not switch the terminal into ring condition (See 4.10.17).

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3.8.14.6 Break-in reset. - Operation of the RESET switch of the terminal receiving a break-in signal shall place the terminal in standby condition (steady mark signal on KEYING and REC jacks and carrier suppression circuit activated) deactivating the audible alarms at both terminals (See 4.10.18).

3.8.15 Alarm circuit. -

3.8.15.1 20 Hz alarm. - An audible signal shall sound on the terminal's panel mounted loudspeaker when a 20 Hz signal is received, and the VF-20 switch is in the 20 position. The maximum signal strength required to sound the alarm shall not exceed 45 volts peak-to-peak. (See 4.10.19)

3.8.15.2 Vf alarm. - An audible signal shall sound on the terminal's panel mounted loudspaker when a 1232.5 Hz (space) signal is received, and the VF-20 switch is in the VF position. The signal strength required to sound the alarm shall be 0 dbm to - 45 dbm. A space signal duration of less than 0.5 second shall not sound the alarm. The audible alarm shall continue as long as the vf signal is received. (See 4.10.20)

3.8.16 Signalling Circuit.

3.8.16.1 20 Hz signal output. - The terminal shall transmit a 20 ± 3 Hz, 90 ± 10% volt (180 volts peak-to-peak) signal into a 1500 -j 1500 ohm load when the ring switch is operated with the VF 20 switch in the 20 position. (See 4.10.21)

3.8.16.2 Vf ring signal output. - The terminal shall transmit a 1232.5 Hz (space), 0 dbm + 4 db - 2 db signal into a 600 ohm resistive load when the RING switch is operated with the VF 20 switch in the VF position (See 4.10.22).

3.8.17 Push to talk radio operation.

3.8.17.1 Two wire operation. - With the 4W-2W-TEL switch in the 2W position, placing the NORM-REC-SEND switch in the SEND position shall provide a dc closure across the 2W-4WR terminals (See 4.10.23).

3.8.17.2 Four wire operation. - With the 4W-2W-TEL switch in the 4W position, placing the NORM-REC-SEND switch in the SEND position shall provide a dc closure across the CONT terminals (See 4.10.23)

3.8.18 Home copy. - With the 4W-HC terminals strapped together, a teletypewriter message transmitted in four wire operation shall appear on the KEYING and REC jacks (See 4.10.24).

3.8.19 Operational. - Terminal, Telegraph TH-22( )/TG shall provide evidence of satisfactory operation when subjected to the operational inspection of 4.10.25.

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3.9 Electrical requirements for Filter Assembly, Electrical F-316( )/U. -

3.9.1 Insertion loss. - The insertion loss of the filter between TP and TG terminals shall be as shown by Table I (See 4.11.1).

3.9.2 Band pass. - The insertion loss of the filter between TG and LINE terminals shall be as shown by Table I (See 4.11.2).

3.9.3 Band stop. - The insertion loss of the filter between TP and LINE terminals shall be as shown by Table I (See 4.11.3).

Table I. - Frequency Response, F-316( )/U

Connection	Frequency Range (Hz)	Attenuation (db)
TP-TG	0-3500	80 minimum
TG-LINE	0-1000	80 minimum
	1180-1225	6 maximum
	1225-1325 (Note 1)	4 maximum
	1325-1600	6 maximum
	1600-3500	80 minimum
TP-LINE	0-1000	4 maximum
	1180-1370	80 minimum
	1600-3500	4 maximum

Note 1. - Mid-frequency shall be 1275 Hz. The attenuation at 1232.5 Hz and 1317.5 Hz shall be within 1 db of each other.

3.9.4 Residual effects. - The filter band stop section shall pass a 20 Hz, 100ma signal without damage or effect on the characteristics of the filter (See 4.11.4).

3.9.5 Cross talk. - A -40dbm vf teletypewriter signal passing through the TG to LINE path of the filter shall not be degraded or distorted by a +5dbm speech signal simultaneously passing through the TP to LINE path (See 4.11.5).

3.9.6 Dielectric strength. - There shall be no voltage breakdown from each terminal to case and between LINE terminals when a 1200 volt dc potential is applied for not less than 3 seconds nor more than 10 seconds (See 4.11.6).

3.9.7 Impedance. - The filter shall present an impedance of 600 ohms ± 10 percent across the LINE, across the TP and across the TG terminals. (See 4.11.7).

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3.9.8 Balance. - The dc resistance measured between the upper LINE and TP terminals shall be within ± 5 percent of the resistance measured between the lower LINE and TP terminals (See 4.11.8).

3.10 Electrical requirements for Converter, Telegraph-Telephone Signal CV-425( )/U. -

3.10.1 Power sources. - The converter shall operate from power sources of 115 or 230 ± 10 percent volts ac, 50 to 60 ± 3 Hz (See 4.12.1).

3.10.2 Vf output frequencies. - The converter ring send circuit shall generate the following output frequencies (See 4.12.2):

TP TG Switch Position	Frequency (Hz)
TG	1232.5 ± 10
TP	1600 ± 10

3.10.3 Vf output level - The vf output level of the converter into a 600 ohm resistive load shall be 0dbm ± 2db when triggered by a 20 Hz, 90 volt (180 volt peak to peak) input (See 4.12.2).

3.10.4 20 Hz output. - The converter shall transmit a 20 ± 3 Hz, 90 ± 10% volt (180 volts peak to peak) signal into a 1500 - j1500 ohm load across the LOOP binding posts when triggered by a vf input as shown by Table II (See 4.12.3).

3.10.5 Vf sensitivity. - The vf signal input level required to trigger the converter 20 Hz output shall be as shown by Table II. Vf signal levels below -55 dbm shall not key the 20 Hz output (See 4.12.3).

Table II. - CV-425( )/U Vf Sensitivity

TP-TG Switch Position	Input Frequency(Hz)	SENSITIVITY Switch Position	Maximum Input to Trigger 20 Hz Output
TG	1195	LOW	+2dbm
	1232.5	LOW	-30dbm
	1232.5	HICH	-45dbm
	1260	LOW	+2dbm
TP	1570	LOW	+2dbm
	1600	LOW	-30dbm
	1600	HICH	-45dbm
	1630	LOW	+2dbm

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3.10.6 20 Hz sensitivity. - The 20 Hz signal input level required to trigger the converter vf output shall be not more than 25 volts (50 volts peak to peak) (See 4.12.4).

3.10.7 Vf guard circuit. - The converter time delay guard circuit shall prevent triggering of the 20 Hz alarm until the incoming vf ring signal has persisted for 0.5 to 2.0 seconds (to prevent triggering of the 20 Hz ring signal by spurious signals or vf teletypewriter signals)(See 4.12.5).

3.10.8 Signal loss. - The signal loss through the converter shall be less than 0.5 db to a 0dbm to -40dbm 1000Hz signal received on the 2W 4W-S LINE binding posts with a 600 ohm resistive load connected across the 2W 4W-S LOOP binding posts. (See 4.12.6)

3.10.9 Distortion. - The total peak distortion introduced by the converter into a 0 dbm 1000 Hz signal passing through it shall not exceed 1 percent (See 4.12.7).

3.10.10 Line input impedance. - The input impedance of the converter measured across the LINE binding posts shall be not less than 8000 ohms when idle or when receiving a 200 to 3500 Hz 0dbm signal from the line. The dc impedance across the LINE binding posts shall be zero ohms when the converter is idle (See 4.12.8).

3.10.11 Line output impedance. - The output impedance of the converter across the LINE binding posts shall be 600 $\pm$ 10% ohms when transmitting a 200 to 3500 Hz signal. (See 4.12.9)

3.10.12 Operational. - Converter, Telegraph-Telephone Signal CV-425( )/U shall provide evidence of satisfactory operation when subjected to the operational inspection of 4.12.10.

3.11 Service conditions. - The equipment shall meet the following service conditions.

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3.11.1 Temperature. -

3.11.1.1 Low. - During testing as specified in 4.13.1, the equipment shall meet the following requirements:

Terminal, Telegraph TH-22( )/TG

3.8.2	Vf output frequencies
3.8.3	Vf output level
3.8.4	Distortion
3.8.9	DC loop output, internal power
3.8.11	Receive circuit sensitivity
3.8.14.1	Break-in transmission
3.8.14.1.1	Break-in signal level
3.8.15.1	20 Hz alarm
3.8.15.2	Vf alarm
3.8.16.1	20 Hz signal output

Filter Assembly, Electrical F-316( )/U

3.9.1	Insertion loss
3.9.2	Band pass
3.9.3	Band stop
3.9.5	Cross talk

Converter, Telegraph-Telephone Signal CV-425( )/U

3.10.2	Vf output frequencies
3.10.3	Vf output level
3.10.4	20 Hz output
3.10.5	Vf sensitivity
3.10.6	20 Hz sensitivity
3.10.8	Signal loss
3.10.9	Distortion

3.11.1.2 High. - During testing as specified in 4.13.2, the equipment shall meet the following requirements:

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3.11.1.2 con't

Terminal, Telegraph TH-22( )/TG

3.8.2	Vf output frequencies
3.8.3	Vf output level
3.8.4	Distortion
3.8.9	DC loop output, internal power
3.8.11	Receive circuit sensitivity
3.8.14.1	Break-in transmission
3.8.14.1.1	Break-in signal level
3.8.15.1	20 Hz alarm
3.8.15.2	Vf alarm
3.8.16.1	20 Hz signal output

Filter Assembly, Electrical F-316( )/U

3.9.1	Insertion loss
3.9.2	Band pass
3.9.3	Band stop
3.9.5	Cross talk

Converter, Telegraph-Telephone Signal CV-425( )/U

3.10.2	Vf output frequencies
3.10.3	Vf output level
3.10.4	20 Hz output
3.10.5	Vf sensitivity
3.10.6	20 Hz sensitivity
3.10.8	Signal loss
3.10.9	Distortion

3.11.2 Humidity (moisture-resistance). - During testing as specified in 4.14, the equipment shall meet the following requirements:

Terminal, Telegraph TH-22( )/TG

3.8.2	Vf output frequencies
3.8.3	Vf output level
3.8.4	Distortion
3.8.9	DC loop output, internal power
3.8.11	Receive circuit sensitivity
3.8.14.1	Break-in transmission
3.8.14.1.1	Break-in signal level
3.8.15.1	20 Hz alarm
3.8.15.2	Vf alarm
3.8.16.1	20 Hz signal output

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3.11.2 con't

Filter Assembly, Electrical F-316( )/U

3.9.1	Insertion loss
3.9.2	Band pass
3.9.3	Band stop
3.9.5	Cross talk

Converter, Telegraph-Telephone Signal CV-425( )/U

3.10.2	Vf output frequencies
3.10.3	Vf output level
3.10.4	20 Hz output
3.10.5	Vf sensitivity
3.10.6	20 Hz sensitivity
3.10.8	Signal loss
3.10.9	Distortion

3.11.3 Altitude (elevation). - During testing as specified in 4.15, the equipment shall meet the operational requirements of 3.8.19 (TH-22( )/TG) and 3.10.12 (CV-425( )/U).

3.11.4 Leakage (immersion). - The equipment shall show no evidence of leakage after it is tested as specified in 4.16. The leakage test for Filter Assembly, Electrical F-316( )/U shall be conducted with the cover removed.

3.11.5 Vibration, bounce and shock. -

3.11.5.1 Vibration, resonance search. - The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz. Vibration test is not required on Filter Assembly, Electrical F-316( )/U (See 4.17).

3.11.5.2 Bounce, loose cargo. - Following testing as specified in 4.18, the equipment shall meet full specification performance.

3.11.5.3 Shock, bench-handling. - Following testing as specified in 4.19, the equipment shall meet the operational requirements of 3.8.19 (TH-22()/TG) and 3.10.12 (CV-425( )/U).

3.11.5.4 Shock, drop. - Following testing as specified in 4.20, the equipment shall meet full specification performance.

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3.12 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.13 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Cleaning prior to welding.
Process for welding.
Cleaning after welding.
Controls
Self-tapping screws.

(b) In this specification:

3.2 Construction

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) First article inspection. (See 4.3) Does not include preparation for delivery.

(b) Inspection covered by subsidiary documents. (See 4.4)

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4.2 con't

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.21)

4.3 First article. - Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. - The contractor shall furnish two first article units of each equipment specified in the contract or purchase order.

4.3.2 First article inspection. - The first article inspection shall consist of inspections specified in Table III, and shall be performed in the order specified therein.

4.3.3 First article data. - The first article test plan and test report(s) shall be as required in the contract or purchase order.

Table III. - First Article Inspection

Inspections	Reqt. Para.	Test Para.	Unit 1	Unit 2
1. Inspections covered by subsidiary document		4.4	See Note d	
2. Bounce preconditioning	3.7	4.6.1	See Note d	
3. Electrical preconditioning	3.7	4.6.2	See Note d	
4. Group A inspection	See Table IV		See Note d	
5. Group B inspection	See Table V		See Note d	
6. Group C inspection				
Low temperature	3.11.1.1	4.13.1	1	
High temperature	3.11.1.2	4.13.2	2	
Altitude	3.11.3	4.15		1
Vibration	3.11.5.1	4.17		2
Humidity	3.11.2	4.14		3
Bounce, loose cargo	3.11.5.2	4.18	3	
Shock, bench-handling	3.11.5.3	4.19	4	
Shock, drop	3.11.5.4	4.20		4

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NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The implementation of inspections covered by 6 shall be expedited to satisfy contract delivery requirements.

c. The numbers in the unit columns in the table covered by 6 specify the order of inspections for the indicated unit.

d. To be performed on all units.

4.4 Inspections covered by subsidiary documents: The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where required
Construction	3.2
Parts, materials and processes	3.3
Finish	3.4
Marking	3.5

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the preconditioning and inspections specified in 4.4, 4.6 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general levels, and AQL's indicated in Table IV.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government, except that the operational in 4.10.25 and 4.12.10 shall be last.

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Table IV. - Group A inspection

Inspection	Req. Para	Insp. Para	AQL Major	AQL Minor
Visual and Mechanical	3.13	4.8	1%	4%
Electrical, TH-22( )/TG				
Power sources	3.8.1	4.10.1		(Note 2)
Transients	3.8.1.1	4.10.1.1		
Polarization	3.8.1.2	4.10.1.2		
Vf output frequencies	3.8.2	4.10.2		
Vf output level	3.8.3	4.10.3		
Distortion	3.8.4	4.10.4		
Signal delay	3.8.5	4.10.5		
Carrier suppression				
Normal signal	3.8.6.1	4.10.6		
Send signal	3.8.6.2	4.10.6		
Receive signal	3.8.6.3	4.10.6		
Send circuit impedance	3.8.7	4.10.7		
Send jack current	3.8.8	4.10.8		
DC loop output, internal power	3.8.9	4.10.9		
DC loop output, external power	3.8.10	4.10.10		
Receive circuit sensitivity	3.8.11	4.10.11		
Receive circuit impedance	3.8.12	4.10.12		
Threshold circuit	3.8.13	4.10.13	4.0 dphu for the group Combined (Note 1)	
Break-in circuit				
Break-in transmission	3.8.14.1	4.10.14		—
Break-in signal level	3.8.14.1.1	4.10.14		
Break-in circuit lock out	3.8.14.1.2	4.10.14		
Break-in interference	3.8.14.2	4.10.15		
Break-in reception	3.8.14.3	4.10.16		
Break-in sensitivity	3.8.14.4	4.10.16		
Break-in time delay	3.8.14.5	4.10.17		
Break-in reset	3.8.14.6	4.10.18		
Alarm circuit				
20 Hz alarm	3.8.15.1	4.10.19		
Vf alarm	3.8.15.2	4.10.20		
Signalling circuit				
20 Hz signal output	3.8.16.1	4.10.21		
Vf ring signal output	3.8.16.2	4.10.22		
Push to talk radio operation				
Two wire operation	3.8.17.1	4.10.23		
Four wire operation	3.8.17.2	4.10.23		
Home copy	3.8.18	4.10.24		
Operational	3.8.19	4.10.25	1%	—

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con't Table IV. - Group A inspection

Inspection	Req. Para	Insp. Para	AQL Major	AQL Minor
Electrical, F-316( )/U				
Insertion loss	3.9.1	4.11.1		(Note 2)
Band pass	3.9.2	4.11.2		
Band stop	3.9.3	4.11.3		
Residual effects	3.9.4	4.11.4	1.5 dphu for the group combined (Note 1)	—
Cross talk	3.9.5	4.11.5		
Dielectric strength	3.9.6	4.11.6		
Impedance	3.9.7	4.11.7		
Balance	3.9.8	4.11.8		
Electrical, CV-425( )/U				
Power sources	3.10.1	4.12.1		
Vf output frequencies	3.10.2	4.12.2		
Vf output level	3.10.3	4.12.2		
20 Hz output	3.10.4	4.12.3	2.5 dphu for the group combined (Note 1)	—
Vf sensitivity	3.10.5	4.12.3		
20 Hz sensitivity	3.10.6	4.12.4		
Vf guard circuit	3.10.7	4.12.5		
Signal loss	3.10.8	4.12.6		
Distortion	3.10.9	4.12.7		
Line input impedance	3.10.10	4.12.8		
Line output impedance	3.10.11	4.12.9		
Operational	3.10.12	4.12.10	1%	—

Notes:

1. - Defects per hundred units.
2. - Defects in all characteristics other than visual and mechanical are major.

4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table V and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B AQL shall be 4.0 percent defective and the inspection level shall be S-4.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

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Table V. - Group B inspection

Inspection	Req Para	Insp Para	AQL
Interchangeability	3.6	4.9	4.0%
Leakage (immersion)	3.11.4	4.16	4.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group inspection. - Two sample units shall be selected at random for all inspection in Table III every two months or from every 500 units or fraction thereof produced which ever occurs first. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within Group C. - Group C inspection shall be performed in accordance with Table III.

4.5.3.3 Group C failures. - Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group B and Group A inspection, in that order.

4.6 Preconditioning. - (See 3.7)

4.6.1 Bounce. - The unit shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute.

4.6.2 Electrical. - Following the bounce preconditioning of 4.6.1, each Terminal, Telegraph TH-22( )/TC and Converter, Telegraph-Telephone Signal CV-425( )/U shall be conditioned for forty-eight hours by applying 110% of rated input voltage (126.5 vac). In addition, any unit which experiences electrical failure during production testing shall be resubjected to the forty-eight hour electrical preconditioning after repair or rework and prior to resumption of testing.

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4.7 Equipment verification review. - The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on order against the drawing cited for construction (Para 3.2). The EVR shall consist of the following:

4.7.1 An audit to establish that "as-built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

4.7.2 An audit to establish that "as-built" configuration of unit, assembly, module, or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.7.3 The contractor shall compile an EVR report(s) containing the following information:

(a) Identification of unit, assembly, module, or part and details proving compliance with 4.7.1 and 4.7.2 above.

(b) Discrepancies noted.

4.7.4 The contractor shall make available to the Government copies of his EVR report(s) no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

(a) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(b) As required, responsible personnel from each functional department available for discussions in their respective areas.

(c) Adequate administrative support for the EVR.

4.8 Visual and mechanical inspection of equipment. - Equipment shall be examined for the defects listed in MIL-STD-252, for the requirements of MIL-P-11268 and for conformance to the construction drawings. (See 3.13)

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4.9 Inspection for dimensional interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.10 Electrical tests for Terminal, Telegraph TH-22( )/TG. -

4.10.1 Power sources test. - Connect the terminal as shown by Figures 2 and 3 for ac and dc power source tests respectively. Set the test message generator to transmit undistorted test message at 100 wpm. With the power source adjusted to each of the setting shown by Table VI, observe whether perfect copy (See 6.4) is produced by the teletypewriter page printer to determine conformance to 3.8.1.

Table VI. - TH-22( )/TG Power source test inputs

Input Voltage (Volts)		Frequencies (Hz)
22	dc	NA
30	dc	NA
104.5	ac	47, 60 and 63
126.5	ac	47, 60 and 63
207	ac	47, 60 and 63
253	ac	47, 60 and 63

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4.10.1.1 Transients test. - Connect the terminal as shown by Figure 3. Set the test message generator to transmit undistorted test message at 100 wpm. Following each of the transient impulses specified by 3.8.1.1, observe whether the teletypewriter page printer prints perfect copy (See 6.4) to determine conformance to 3.8.1.1.

4.10.1.2 Polarization test. - Connect the terminal as shown by Figure 4. The resistance measured across pins A and B of connector 1J5 shall be 10± 20 % ohms with the positive ohmmeter lead connected to pin A to verify conformance to 3.8.1.2.

4.10.2 Output frequency test. - Connect the terminal as shown by Figure 5. With the test message generator adjusted to transmit steady mark, steady space and alternate mark and space at 37.5 Hz, observe the terminal output frequencies to determine conformance to 3.8.2.

4.10.3 Output level test. - Connect the terminal as shown by Figure 5. Measure the output levels of the mark and space signals to determine conformance to 3.8.3.

4.10.4 Distortion test. - Connect the terminal as shown by Figure 6. With the test message generator transmitting undistorted test message at 60, 75, 100 and 200 wpm, measure the distortion at the PLC jack to determine conformance of the terminal to 3.8.4. The test is to be conducted with the attenuator adjusted to provide a -3dbm and a -45dbm signal at the 2W 4WF binding posts at each teletypewriter speed.

4.10.5 Signal delay test. - Connect the terminal as shown by Figure 7. Measure the time delay between transmission of transitions by the test message generator and by the terminal to determine conformance to 3.8.5.

4.10.6 Carrier suppression test. - Connect the terminal as shown by Figure 8. With the NORM REC SEND switch in the NORM position, measure the time delay between closing at S1 and suppression of the vf output, and the time delay between opening of S1 and resumption of vf output to determine conformance to 3.8.6.1. With the NORM REC SEND switch in the REC and the SEND positions, measure the vf output level to determine conformance to 3.8.6.2 and 3.8.6.3.

4.10.7 Send circuit impedance test. - Connect the terminal as shown by Figure 9. Set the 4W 2W TEL Switch to 4W and the NORM REC SEND switch to SEND so that a 1232.5 ± 3 Hz (space) signal is transmitted by the terminal. Measure the output level of the transmitted signal with the 600 ohm resistor both in and out of the circuit. A difference in output level of the loaded and unloaded circuit between 5.1 db and 6.4 db verifies compliance with 3.8.7.

4.10.8 Send jack current test. - Connect the terminal as shown by Figure 10. Measure the current to determine conformance to 3.8.8.

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4.10.9 DC loop output, internal power, test. - Connect the terminal as shown by Figure 11 except that the attenuator is not required. Set the test message generator to steady mark and to steady space, and measure the terminal output voltage and current at the REC jack to verify conformance to 3.8.9. Measurements shall be performed with the variable resistance set at 100 ohms and at 1000 ohms. Mark output current shall also be measured with the variable resistance set at zero ohms (shorted), and the fact that the terminal was not damaged by the short circuit shall be determined by repeating the distortion test of 4.10.4.

4.10.10 DC loop output, external power, test. - Connect the terminal as shown by Figure 12 except that the oscilloscope, frequency counter and two telephones TA-312( )/U are not required. Adjust the variable resistor to place a total load of 1000 ohms across the KEYING jack. Adjust the teletypewriter page printer to supply 60 ma loop current connected to the terminal KEYING jack. Adjust the attenuator to supply the test message at levels of 0 dbm and -45 dbm and observe whether perfect copy (See 6.4) is produced by the teletypewriter page printer to determine conformance to 3.8.10.

4.10.11 Receive circuit sensitivity test. - Connect the terminal as shown by Figure 11 except that the milliammeter is not required. Set the test message generator to alternate marks and spaces at the rate of 37.5 Hz. With the attenuator adjusted to provide signal levels of 0 dbm and -45 dbm at the 2W 4WR binding posts and the variable resistor set at 0 ohms and 1000 ohms, observe the receive circuit output to determine conformance to 3.8.11.

4.10.12 Receive circuit impedance test. - Connect the terminal as shown by Figure 13. With the oscillator set at 1000 Hz and at 2000 Hz, measure the voltage across the 2W 4WR binding posts. Disconnect the terminal from the oscillator, and connect the decade resistor to the oscillator. Adjust the decade resistor to give the same voltage as was measured across the 2W 4WR binding posts. The resistance set on the decade resistor is the input impedance of the terminal. (See 3.8.12)

4.10.13 Threshold circuit tests. - Connect the terminal as shown by Figure 12 except that the two telephones TA-312( )/U are not required. Apply the test message to the unit under test with the high-low sensitivity strap in the high position and the attenuator adjusted so that the signal level delivered to unit under test is 0 dbm. Note whether legible copy is received on the teletypewriter and whether the THRESHOLD lamp is out. Adjust signal level to -45 dbm and again note legible copy and THRESHOLD lamp out. Reduce signal level until teletypewriter remains in closed (steady mark) condition and THRESHOLD lamp lights; measure level to determine conformance to 3.8.13. Return level to 0 dbm and note that teletypewriter resumes operation and the THRESHOLD lamp goes out. Repeat test with sensitivity strap in low position and the lower signal level at -22 dbm. Measure the time delay between reduction of the input signal level to below the threshold level and appearance of steady mark at the REC jack to determine conformance with the time delay requirement of 3.8.13.

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4.10.14 Break-in transmission signal level and lock out tests. - Connect the terminal as shown by Figure 12 but with the attenuator switched out of the circuit. Set the message generator to transmit undistorted test message. Operate the RESET switches of both terminals and note that neither audible alarm is operating. Operate the BREAK IN switch of the unit under test, and note that the audible alarm of the unit not under test sounds. Note that until the BREAK IN Switch is released, the teletypewriter runs open. Measure the output frequency and level of the unit under test with the BREAK IN switch operated and with and without the BOOST switch operated to determine conformance to 3.8.14.1 and 3.8.14.1.1. Release the BREAK IN switch; note that both audible alarms sound and that the teletypewriter is in closed (steady mark) condition (See 3.8.14.1.2). Operate the RESET switch of the unit not under test; note that both audible alarms cease to sound and that the teletypewriter resumes printing the test message. Operate the BREAK IN switch with the break-in disable strap installed to determine whether transmission of the break-in signal has been prevented in accordance with 3.8.14.1.

4.10.15 Break in interference test. - During operation of the BREAK IN and RESET switches in 4.10.14, note whether interference is audible on either the two Telephone Sets TA-312( )/U to determine conformance to 3.8.14.2.

4.10.16 Break in reception and sensitivity test. - Connect the terminal as shown by Figure 14 but with the attenuator switched out of the system. Set the message generator to transmit undistorted test message at 100 wpm. Operate the RESET of both terminals, and note that neither audible alarm is operating. Operate the BREAK IN switch of the unit not under test, and note that the audible alarm of the unit under test sounds and that a space (open) signal is present on the SEND jack of the unit under test. Release the BREAK IN switch and measure the vf output frequency and level of the unit under test to determine conformance to 3.8.14.3. Repeat the test with the attenuator set to provide a -30 dbm and a -32 dbm signal to the unit under test to determine conformance to 3.8.14.4.

4.10.17 Break-in time delay test. - During operation of the BREAK IN switch of the unit not under test in 4.10.16, measure the duration of the break-in required to trigger unit under test to ring condition to determine conformance to 3.8.14.5.

4.10.18 Break-in reset test. - Following completion of the tests of 4.10.16, operate the RESET switch of the unit under test. Note whether the audible alarms cease to sound and whether teletypewriter resumes printing the test message to determine conformance to 3.8.14.6.

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4.10.19 20 Hz alarm test. - Connect the terminal as shown by Figure 15 with the attenuator out of the circuit and the amplifier adjusted to 45 volts peak to peak. Set 4W 2W TEL Switch to 4W, close S1 and note whether audible alarm sounds to determine conformance to 3.8.15.1. Note whether audible alarm is turned off by opening S1. Set the 4W 2W TEL switch to 2W and repeat the test.

4.10.20 VF alarm test. - Connect the terminal as shown by Figure 15 with the amplifier out of the circuit and the attenuator connected and adjusted to provide a 0 dbm signal to the terminal. Set 4W 2W TEL switch to 4W, close S1 and note whether the audible alarm sounds to determine conformance to 3.8.15.2. Note that the audible alarm is turned off by opening S1. Repeat at -45 dbm input. Set 4W 2W TEL switch to 2W and repeat the test at 0 dbm and -45 dbm input. Measure the time delay between closing S1 and sounding of the audible alarm to determine conformance to the time delay requirement of 3.8.15.2.

4.10.21 20 Hz signal output test. - Connect the terminal as shown by Figure 16. Operate the RING switch, and measure the output frequency and level to determine conformance to 3.8.16.1.

4.10.22 VF ring signal output test. - Connect the terminal as shown by Figure 5 but with the test message generator removed from the circuit. Operate the RING switch and measure the output frequency and level to determine conformance to 3.8.16.2.

4.10.23 Push to talk tests. - Connect the terminal as shown by Figure 17. The dc resistance measured across the 2W-4WR terminals shall be 100 ± 10% ohms and across the CONT terminals shall be 0 ohms to verify conformance to 3.8.17.1 and 3.8.17.2 respectively.

4.10.24 Home copy test. - Connect the terminal as shown by Figure 7. Observe whether a home copy of a message transmitted in four wire operation is printed by the teletypewriter page printer to determine conformance to 3.8.18.

4.10.25 Operational inspection. - Connect the terminal as shown by Figure 6 except that the distortion analyzer is not required. Set the test message generator to transmit undistorted test message at 100 wpm. Observe whether the teletypewriter prints perfect copy (See 6.4) to determine conformance to 3.8.19. (See 6.6)

4.11 Electrical tests for Filter Assembly, Electrical F-316( )/U

4.11.1 Insertion loss test. - Connect the filter as shown by Figure 18. Measure the signal levels (attenuation) to determine conformance to 3.9.1.

4.11.2 Band pass test. - Connect the filter as shown by Figure 19. Measure the signal levels (attenuation) to determine conformance to 3.9.2.

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4.11.3 Band stop test. - Connect the filter as shown by Figure 20. Measure the signal levels (attenuation) to determine conformance to 3.9.3.

4.11.4 Residual effects test. - Connect the filter as shown by Figure 21. Measure the level of the maximum odd harmonic of 20Hz. The level must be less than -55 dbm to verify conformance to 3.9.4.

4.11.5 Cross talk test. - Connect the filter as shown by Figure 22. Set the oscillator level to +5 dbm at the LINE terminals. Measure the 1275 Hz level at the TG terminals. The level must be less than -55 dbm to verify conformance to 3.9.5.

4.11.6 Dielectric strength test. - Connect the filter as shown by Figure 23. Apply 1200 vdc potential between each terminal and the case and between LINE terminals for four seconds after static capacity is charged. Measure the current flow during the four second period. A measurement of less than one microampere is required to verify conformance to 3.9.6.

4.11.7 Impedance test. - Connect the filter as shown by Figure 24. With the oscillator set at 1275 Hz, measure the voltage across the TP terminals. Disconnect the filter from the oscillator, and connect the decade resistor to the oscillator. Adjust the decade resistor to give the same voltage as was measured across the TP terminals. The resistance set on the decade resistor is the impedance across the TP terminals. Repeat across the TG and LINE terminals to determine conformance to 3.9.7.

4.11.8 Balance test. - Connect the filter as shown by Figure 25. Measure the dc resistance between the upper TP and LINE terminals and between the lower TP and LINE terminals to determine conformance to 3.9.8.

4.12 Electrical tests for Converter, Telegraph-Telephone Signal CV-425()/U

4.12.1 Power sources test. - Connect the converter as shown by Figure 26. Set the 20Hz input level to 180 volts peak to peak. With the power source adjusted to each of the settings shown by Table VII, the vf output frequencies and level must be in accordance with 3.10.2 and 3.10.3 respectively to verify conformance to 3.10.1.

Table VII. - CV-425( )/U Power Supply Test Inputs

Input Voltage (Volts ac)	Frequencies (Hz)
104.5	47, 60 and 63
126.5	47, 60 and 63
207	47, 60 and 63
253	47, 60 and 63

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4.12.2 Vf output and level tests. - Connect the converter as shown by Figure 26. Set the 20 Hz input level to 180 volts peak to peak and the power source to 115 volts, 60 Hz. Measure the vf output frequencies and levels to determine conformance to 3.10.2 and 3.10.3.

4.12.3 20Hz output and vf sensitivity tests. - Connect the converter as shown by Figure 27. Adjust the input frequency and level to each of the settings shown by Table II. Measure the output frequency and level of the converter to determine conformance to 3.10.4 and 3.10.5.

4.12.4 20Hz sensitivity test. - Connect the converter as shown by Figure 26. Set the 20Hz input level at 50 volts peak to peak. With the power source adjusted to 115 volts 60 Hz, the vf output frequency and level must be in accordance with 3.10.2 and 3.10.3 to verify conformance to 3.10.6.

4.12.5 Vf guard circuit test. - Connect the converter as shown by Figure 28. Adjust the oscilloscope to trigger when S-1 is closed. Measure the time delay between closure of S-1 and transmission of a 20Hz signal from the converter to determine conformance to 3.10.7

4.12.6 Signal loss test. - Connect the converter as shown by Figure 29. Adjust input level to Odbm and measure output; adjust input level to -40 dbm and measure output to determine conformance to 3.10.8.

4.12.7 Distortion test. - Connect the converter as shown by Figure 30. Measure the distortion introduced into the signal by the converter to determine conformance to 3.10.9.

4.12.8 Line input impedance test. - Connect the converter as shown by Figure 31. With the oscillator set at 200, 1000 and 3500 Hz measure the voltage across the 2W 4WS and the 4WF LINE binding posts. Disconnect the converter from the oscillator and connect the decade resistor to the oscillator. Adjust the decade resistor to give the same voltage as was measured across the converter binding posts. The resistance set on the decade resistor is the input impedance of the converter (See 3.10.10).

4.12.9 Line output impedance test. - Connect the converter as shown by Figure 32. Measure the output level of the transmitted signal with the 600 ohm resistor both in and out of the circuit. A difference in power level of the loaded and unloaded circuit between 5.6 db and 6.4 db verifies compliance to 3.10.11.

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4.12.10 Operational inspection. - Connect the converter as shown by Figure 26 except that the variable power source is not required. Apply a 20 Hz, 50 volt peak to peak signal to the converter. Measure the vf output frequency and level with the TP TG switch in the TP and TG positions to determine conformance to 3.10.2 and 3.10.3. Connect the converter as shown by Figure 27. Apply 1232.5 and 1600 Hz, -30dbm and -45 dbm signals with the TP TG switch in the TP and TG positions and the SENSITIVITY switch in the LOW and HIGH positions. Measure the 20 Hz output frequency and level to determine conformance to 3.10.4 (See 3.10.12 and 6.6).

4.13 Temperature test. -

4.13.1 Low Temperature. - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storate temperature shall be -80°F for a period of 24 hours, and the operating temperature shall be -40°F (See 3.11.1.1).

4.13.2 High temperature. - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 160°F (See 3.11.1.2).

4.14 Humidity (moisture-resistance) test. - The equipment shall be tested in accordance with Method 507, Procedure II of MIL-STD-810. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test (See 3.11.2).

4.15 Altitude (elevation) test. - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810 (See 3.11.3).

4.16 Leakage (immersion) test. - The equipment, as prepared for field transportation (except that F-316( )/U cover is removed), shall be tested in accordance with Method 512, Procedure I, of MIL-STD-810. In addition to the failure criteria listed in the test method, a continuous stream of bubbles for 30 seconds or longer is evidence of leakage (See 3.11.4).

4.17 Vibration, resonance search test. - The equipment shall be tested in accordance with Method 514, Procedure IX, Part 1 of MIL-STD-810 (See 3.11.5.1).

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4.18 Bounce, loose cargo test. - The equipment shall be tested in accordance with Method 514, Procedure XI, Part 2 of MIL-STD-810 (See 3.11.5.2)

4.19 Shock, bench-handling. - The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810 (See 3.11.5.3).

4.20 Shock, drop test. - The equipment shall be tested in accordance with Method 516, Procedure II, of MIL-STD-810 except that the deight of drop shall be 36 inches (See 3.11.5.4).

4.21 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (See 6.2(c)).

5.1.1 Level A.

5.1.1.1 Cleaning. Terminal, Telegraph-Telephone AN/TCC-29( ) and units of shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Terminal, Telegraph-Telephone AN/TCC-29( ) and units of shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Spare fuses. Each spare fuses shall be placed in the clips provided in the cover.

5.1.1.4.3 Spare lamp. Each spare lamp shall be packaged method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, Type III, secure wrapping with tape conforming to PPP-T-45, Type III.

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5.1.1.4.4 Terminal-Telegraph. TH-22( )/TG. Each Terminal-Telegraph shall be individually packaged Method III as follows: Coil the cable within the case, cushion the end plug with paperboard conforming to PPP-P-291, Type III, secure the lamp packaged as specified in 5.1.1.4.3 and the end plug to the handle with tape conforming to PPP-T-76. Secure the case cover with the fastenings provided. The carrying straps shall be tightly drawn through the brackets and buckle provided. Cushion the unit on all surfaces with cells or pads or both, fabricated of fiberboard conforming to PPP-F-320, Type CF, class weather-resistance variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the suchioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.1.4.5 Converter, Telegraph-Telephone Signal CV-425( )/U. Each converter shall be packaged method III in accordance with the procedure specified in 5.1.1.4.4.

5.1.1.4.6 Filter Assembly, Electrical F-316( )/U. Each filter assembly shall be packaged method III in accordance with the procedure specified in 5.1.1.4.4.

5.1.1.4.7 Terminal, Telegraph-Telephone AN/TCC-29( ). Each terminal telegraph-telephone, unit packaged in accordance with the procedures specified in 5.1.1.4.4 thru 5.1.1.4.6, shall be consolidated within a close-fitting box conforming to PPP-B-636, W5c. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level C. Each Terminal, Telegraph-Telephone AN/TCC-29( ) or individual units shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C, as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents. (See 6.2(c)).

5.2.1 Level A.

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5.2.1.1 Consolidation. A quantity of Terminal-Telegraph TH-22( )/TG or Converter, Telegraph-Telephone Signal CV-425( )/U or Filter Assembly, Electrical F-316( )/U, packaged as specified in 5.1 and bearing the same stock number, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. A quantity of Terminal, Telegraph-Telephone AN/TCC-29( ), packaged as specified in 5.1, or a quantity of items, packed as specified in 5.2.1.1 and bearing the same stock number, shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, the containers used for packaging the Terminal, Telegraph-Telephone AN/TCC-29( ) as specified in 5.1, or the containers, packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers bearing the same stock number shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Consolidation. A quantity of Terminal, Telegraph TH-22( )/TG or Converter, Telegraph-Telephone Signal CV-425( )/U or Filter Assembly, Electrical F-316( )/U, packaged as specified in 5.1, and bearing the same stock number shall be packed as specified in 5.2.1.1.

5.2.2.2 Palletized load. A quantity of Terminal, Telegraph-Telephone AN/TCC-29( ), packaged as specified in 5.1, or a quantity of items, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

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5.2.2.3 Less than palletized load.

5.2.2.3.1 When quantities per destination are less than a pallet load, the containers used to pack the items as specified in 5.2.2.1, shall be reinforced by pressure-sensitive filament tape banding conforming to PPP-T-97, type IV as specified in the appendix of the box specification. No further packing shall be required.

5.2.2.3.2 When quantities per destination are less than a pallet load, a quantity of Terminal, Telegraph-Telephone AN/TCC-29( ), packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Consolidation. A quantity of Terminal-Telegraph TH-22( )/TG or Converter, Telegraph-Telephone Signal CV-425( )/U or Filter Assembly, Electrical F-316( )/U, packaged as specified in 5.1, and bearing the same stock number, shall be packed as specified in 5.2.1.1, except that the fiberboard box shall be class domestic.

5.2.3.2 Palletized load. A quantity of Terminal, Telegraph-Telephone AN/TCC-29( ), packaged as specified in 5.1, or a quantity of items, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load.

5.2.3.3.1 When quantities per destination are less than a pallet load, the containers used to pack the items as specified in 5.2.3.1, shall be used as the shipping container. No further packing shall be required.

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5.2.3.3.2 When quantities per destination are less than a pallet load, a quantity of Terminal Telegraph-Telephone AN/TCC-29( ), packaged as specified in 5.1, shall be packed as specified in 5.2.2.3.2, except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Terminal, Telegraph-Telephone AN/TCC-29( ) is intended primarily for forward area combat application in point-to-point, switched and radio telephone operation of simultaneous teletype and voice communication. The units, i.e., TH-22( )/TC, CV-425( )/U and F316( )/U, may be used in combination or separately to serve the particular needs of the communications system. Terminal, Telegraph-Telephone AN/TCC-29( ) is functionally identical to Terminal, Telegraph-Telephone AN/TCC-14( ).

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number and date of this specification and any amendment thereto.

(b) Type required (See 1.1 and 6.1).

(c) Levels of preservation and packaging, and packing (See Section 5).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) When the rough handling and functional tests are required.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required (See 3.12).

(i) Quantity of tools and running spare parts required (See 3.12).

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/TCC-29W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (See 1.1).

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6.4 Perfect copy. - Perfect copy is defined as ten consecutive error free typed copies of the standard test message: THE QUICK BROWN FOX JUMPED OVER A LAZY DOG'S BACK 1234567890 DTS SENDING.

6.5 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Location of operational inspection. - It is desirable that the operational inspection (4.10.25 and 4.12.10) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

Custodian
Army - EL

Preparing Activity
Army - EL

Project Number 5805-A196

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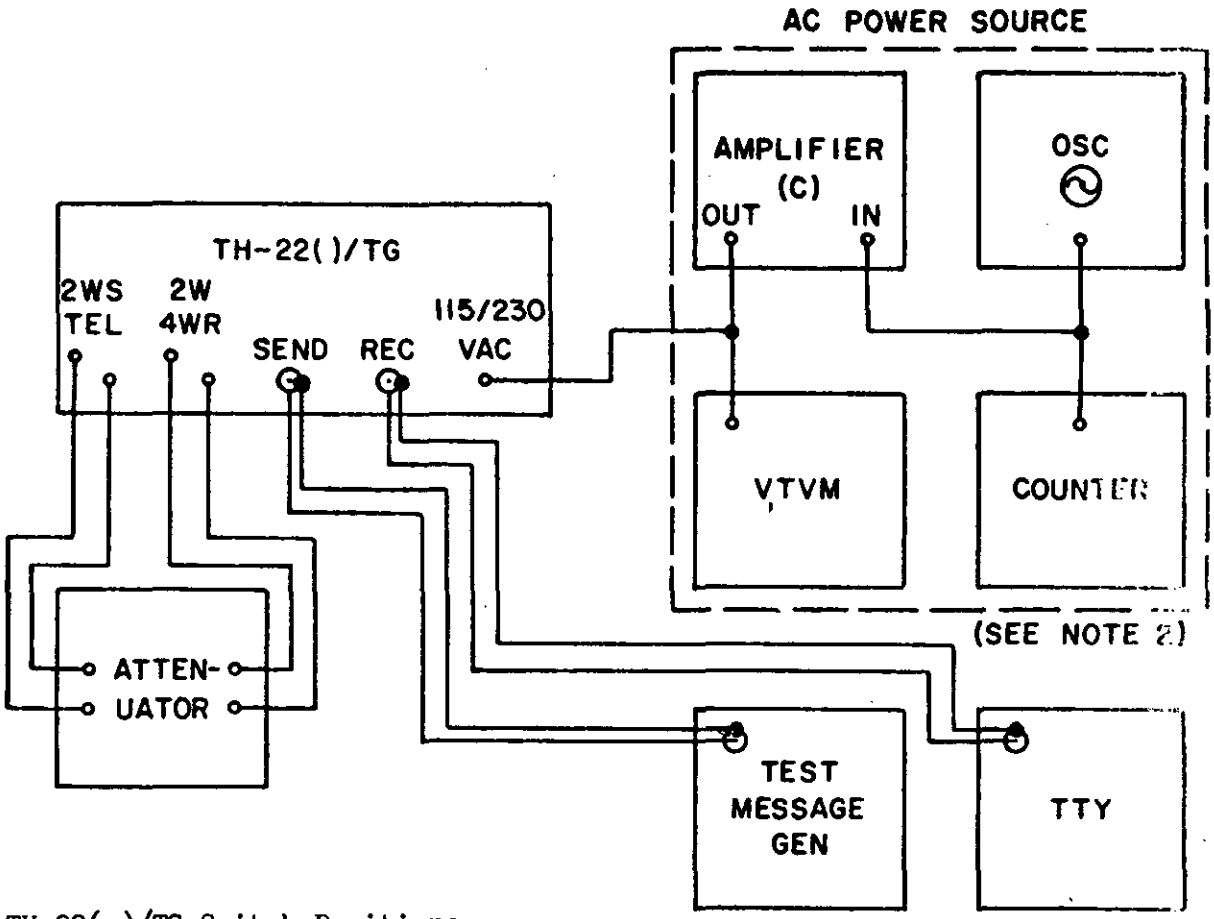
TEST EQUIPMENT REQUIRED, OR EQUAL

Amplifier (C) - Commercial Measurements Laboratory Model N-500A
Amplifier (K) - Kron-Hite Model DCA-10
Attenuator - Daven Model T-690-C
Decade Resistor - General Radio Model 1432P
Distortion Analyzer - Digitech Model 603-3
Frequency Counter - Hewlett Packard Model 5223L
Isolation Transformer (2 Required) - Hewlett Packard Model AC-60B
Multimeter - Simpson Model 260
Adapter - Simpson Model 653
Oscillator - Hewlett Packard Model 200CD
Oscilloscope - Tektronix Model 535A
Plug In - Tektronix Model CA
Telephone (2 Required) - Telephone Set TA-312( )/U
Teletypewriter Page Printer - Teletype Corp Model 33
Test Message Generator - Digitech Model 2102
VTVM (2 Required) - Hewlett Packard Model 400FL
Wave Analyzer - Hewlett Packard Model 302A

Figure 1

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TH-22( )/TG
AC Power Source Test



TH-22( )/TG Switch Positions

115VAC 230VAC 26VDC - 115VAC & 230VAC
ON OFF - ON
NORM REG SEND - NORM
4W 2W TEL-4W
VF 20 - Either
MODE ONE MODE TWO - MODE ONE

Notes

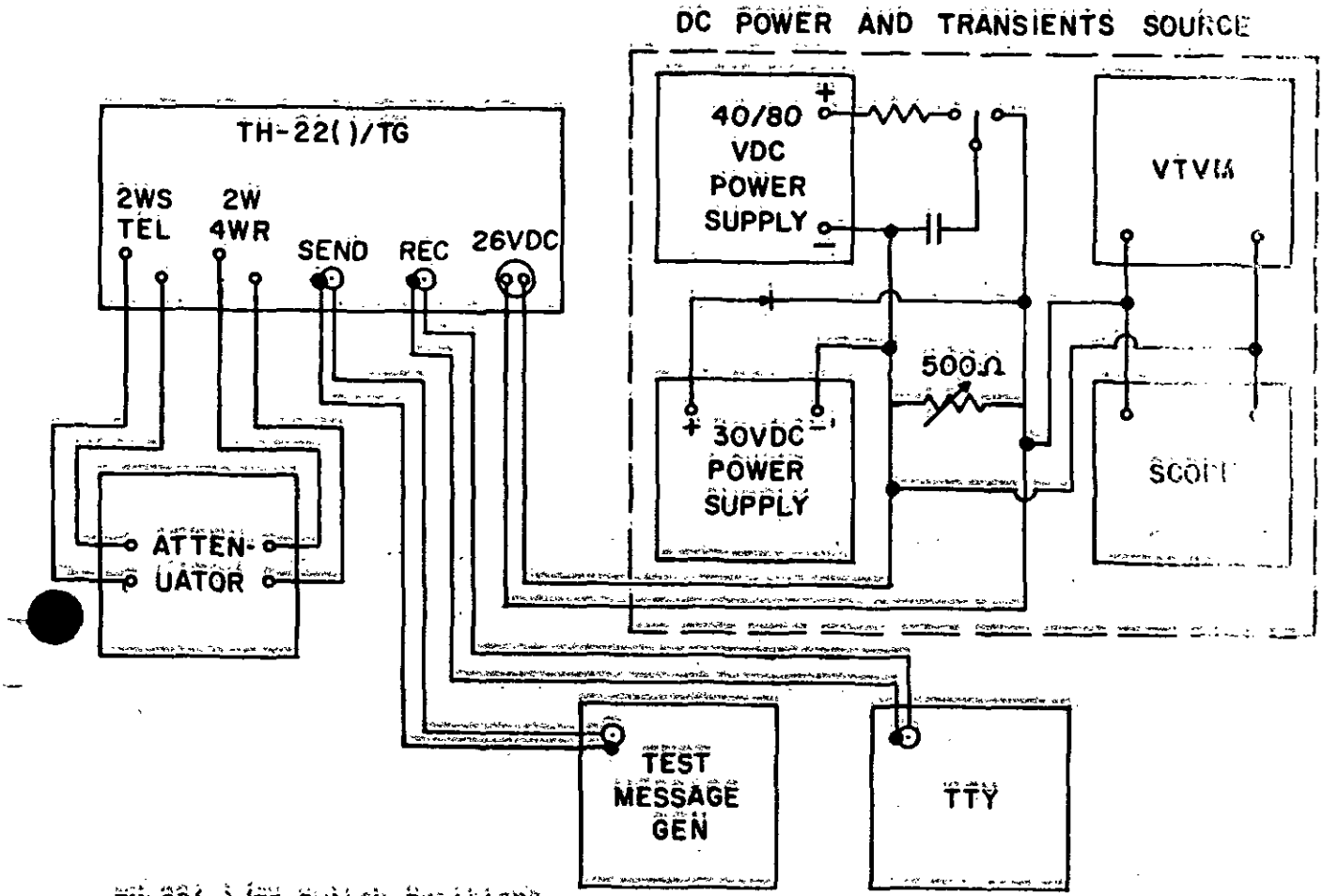
1. See Figure 1 for test equipment descriptions.

2. Power source oscillator frequency and amplifier output level to be adjusted in accordance with power source test paragraph.

Figure 2

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TH-22( )/TG
DC Power Source and Transients Tests



TH-22(_)/TG Switch Positions

115VAC 230VAC 26VDC = 26VDC
ON OFF = ON
NORM REC SEND = NORM
2W 4W TEL=4W
VF 20 = Either
MODE ONE MODE TWO = MODE ONE

Notes

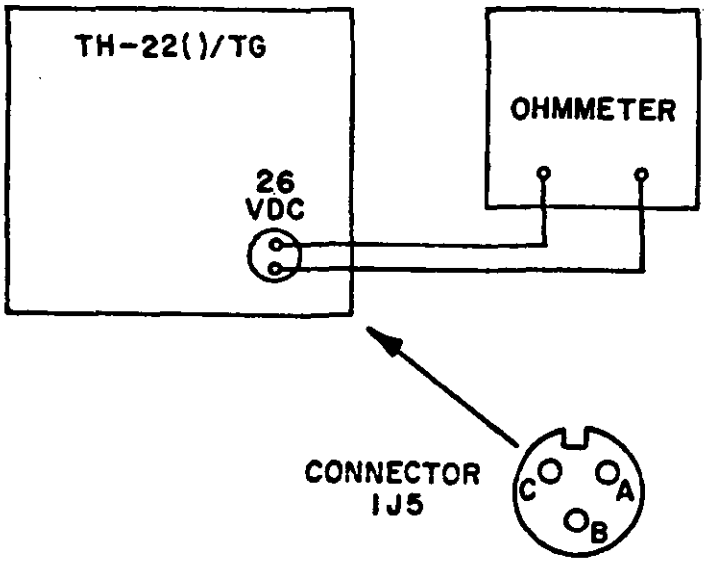
1. See Figure 1 for test equipment descriptions.

2. DC and transient power supplies to be adjusted in accordance with test paragraphs.

Figure 3

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TH-22( )/TG
Polarization Test



TH-22( )/TG Switch Positions

115VAC 230VAC 26VDC - 26VDC
ON OFF - OFF
NORM PEC SEND - Any
4W 2W TEL - Any
VF 20 - Any
MODE ONE MODE TWO - Any

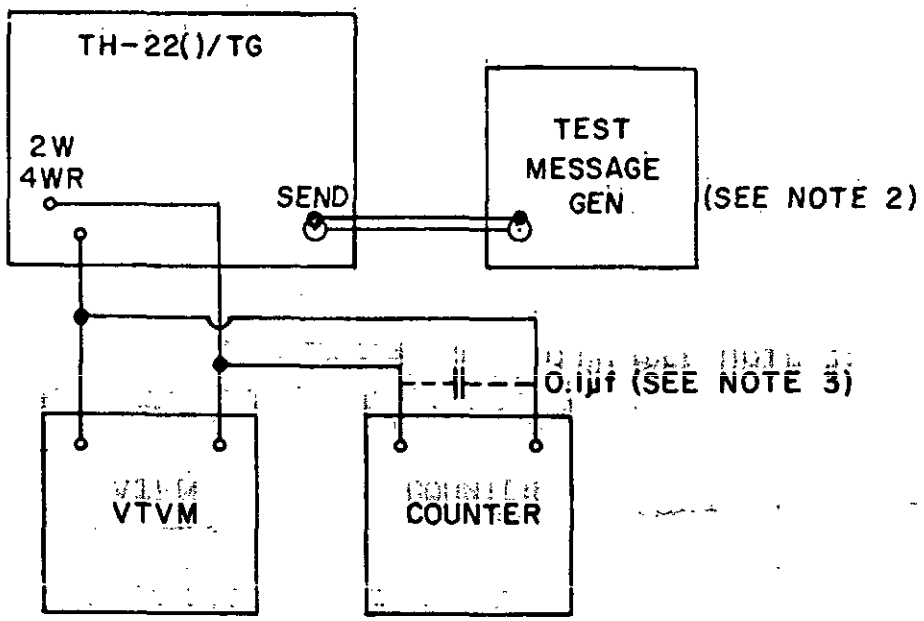
Notes

1. See Figure 1 for test equipment descriptions.

Figure 4

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TH-22( )/TG
Output Frequency and Level and Vf Ring Signal Output



TH-22( )/TG Switch Positions

115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - NORM
4W 2W TEL - 2W
VF 20 - VF
MODE ONE MODE TWO - MODE ONE

Notes

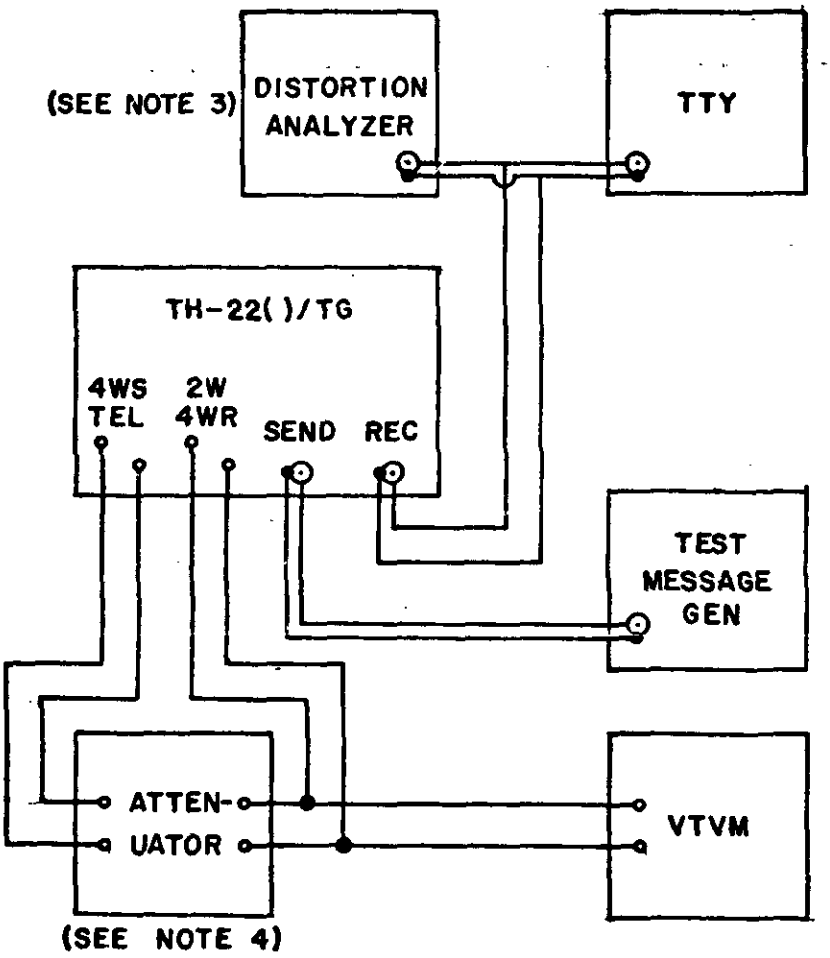
1. See Figure 1 for test equipment descriptions.

2. Test message generator adjusted to steady mark, steady space or dot cycles in accordance with test paragraphs.

3. Capacitor used for average frequency test only.

Figure 5

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Switch Positions

115VAC 230VAC 26 VDC - 115VAC
ON OFF - ON
NOFM REC SEND - NOFM
4W 2W TEL - 4W
VF 20 - Either
MODE ONE MODE TWO - Both (See Note 2)

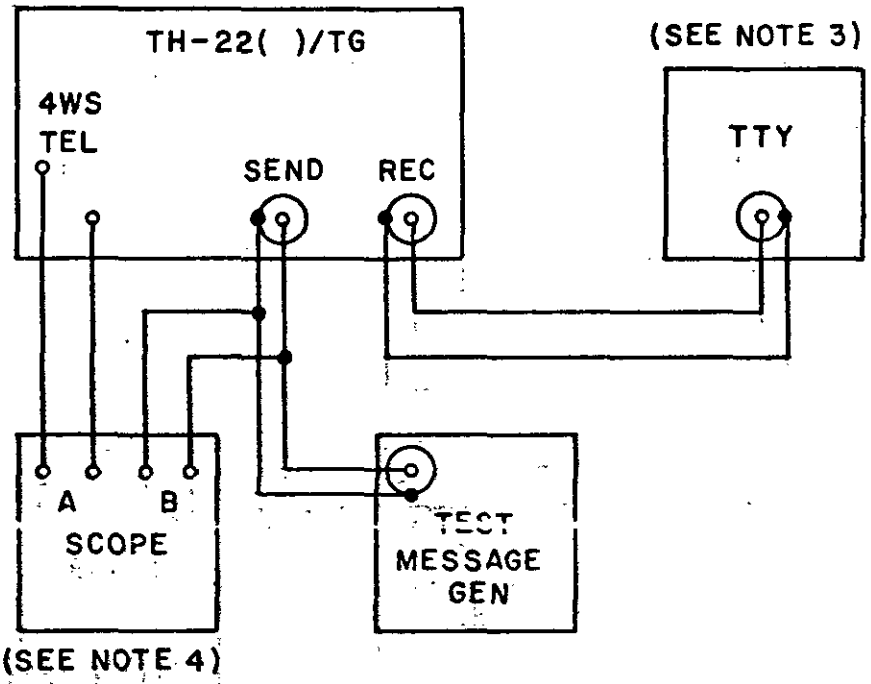
Notes

1. See Figure 1 for test equipment descriptions.
2. Set MODE ONE MODE TWO Switch to MODE TWO for operation at 200 wpm and MODE ONE at 60, 75 and 100 wpm.
3. Distortion analyzer required for distortion test only.
4. Attenuator is adjusted to provide a -45 dbm signal to the 2W 4WR terminals for the operational inspection. For the distortion test, adjust in accordance with test paragraph.

Figure 6

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TH-22( )/TG
Signal Delay and Home Copy Tests



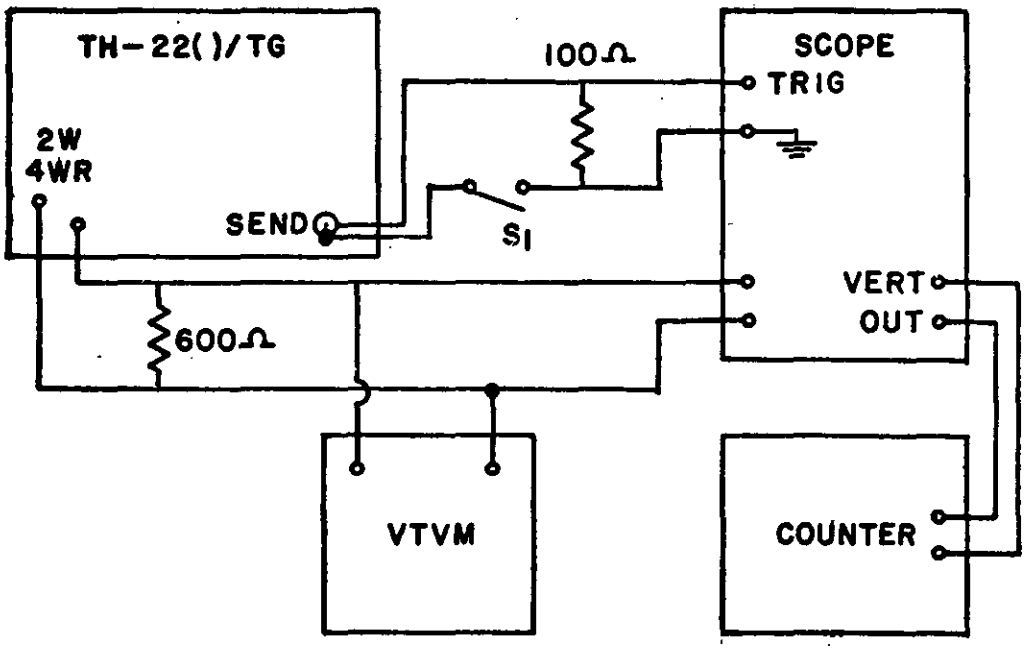
TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - NORM
4W 2W TEL - 4W
VF 20 - Either
MODE ONE MODE TWO - Both

NOTES
1. See Figure 1 for test equipment descriptions.
2. 4W - HC terminals on PC board 1A2 strapped together for home copy test.
3. Teletypewriter page printer required for home copy test only.
4. Oscilloscope required for signal delay test only.

Figure 7

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TH-22( )/TG
Carrier Suppression Test



TH-22( )/TG Switch Positions

115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND -See Note 2
2W 4W TEL - 2W
VF 20 - Either
MODE ONE MODE TWO - MODE ONE

Notes

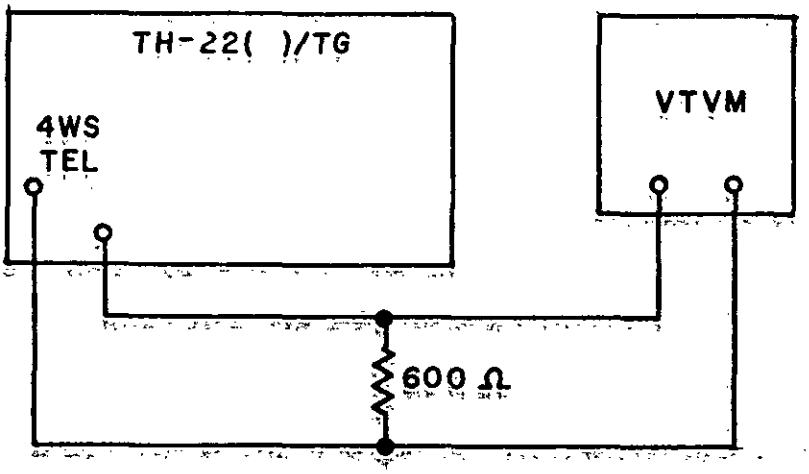
1. See Figure 1 for test equipment descriptions.

2. NORM REC SEND Switch to be positioned in accordance with the applicable test paragraph.

Figure 8

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TH-22( )/TC
Send Circuit Impedance Test



TH-22( )/TG Switch Positions

115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - SEND
4W 2W TEL - 4W
VF 20 - Either
MODE ONE MODE TWO - EITHER

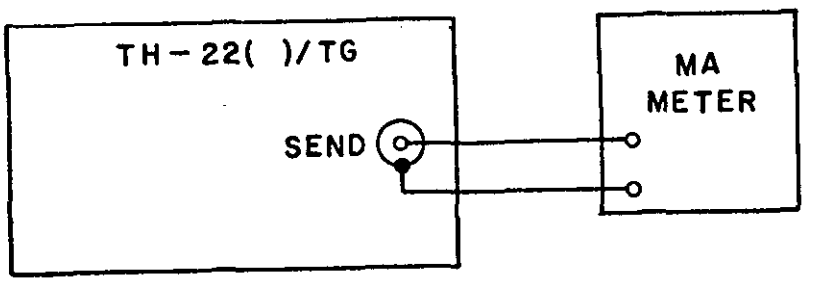
Notes

1. See Figure 1 for test equipment descriptions.

Figure 9

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TH-22( )/TG
Send Jack Current Test



TH-22( )/TG Switch Positions

115VAC 230VAC 26 VDC - 115VAC
ON OFF - ON
NORM SEND RECEIVE - Any
2W 4W TEL - Any
VF 20 - Either
MODE ONE MODE TWO - Either

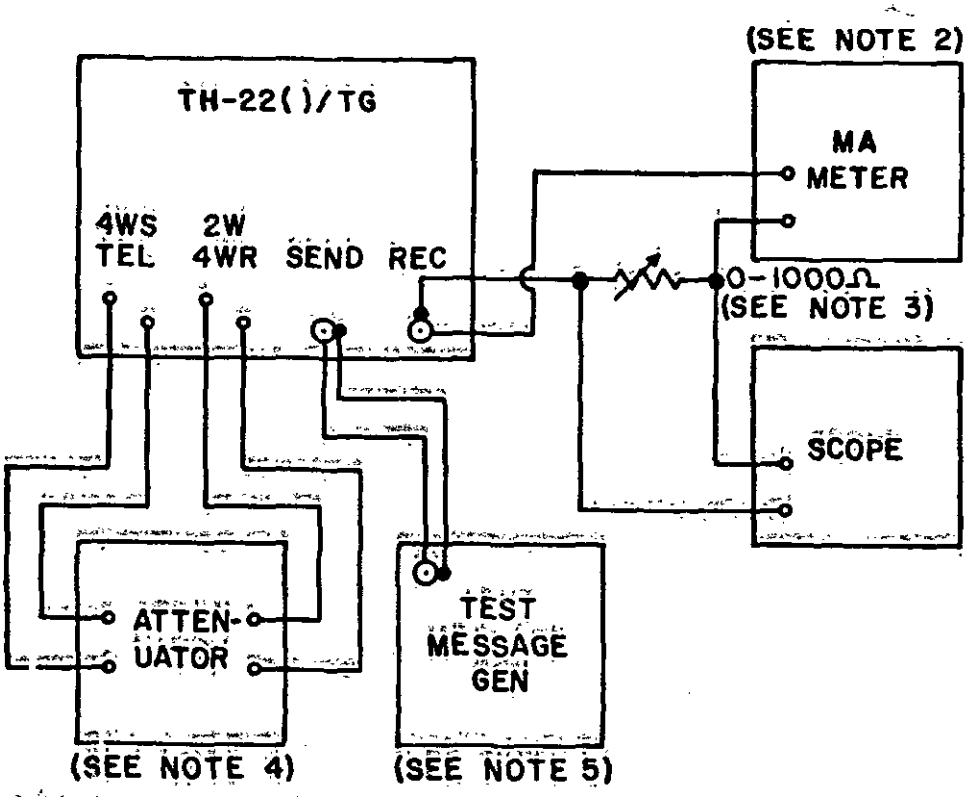
Notes

1. See Figure 1 for test equipment descriptions.

Figure 10

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TH-22( )/TG
DC Loop Output, Internal Power and Receive Circuit Sensitivity Tests



TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - REC
4W 2W TEL - 2W
VF 2O - VF
MODE ONE MODE TWO - MODE ONE

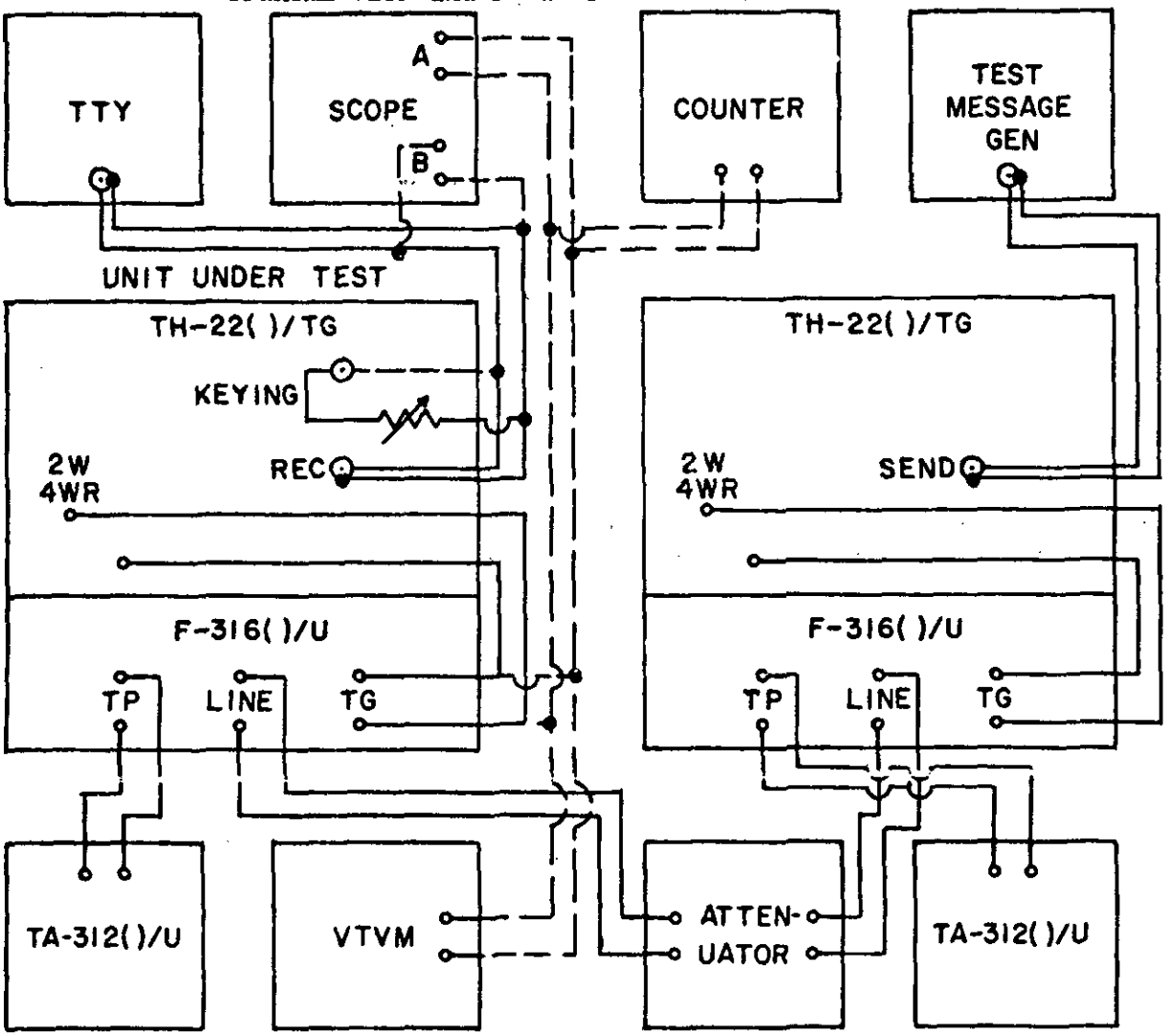
Notes

1. See Figure 1 for test equipment descriptions.
2. Milliammeter required for DC Loop Output, Internal Power Test only.
3. Variable resistance to be adjusted in accordance with appropriate test paragraph.
4. Attenuator required for Receive Circuit Sensitivity Test only. To be adjusted in accordance with test paragraph.
5. Test message generator to be adjusted in accordance with applicable test paragraph.

Figure 11

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TH-22( )/TG
DC Loop Output-External Power, Threshold, Break In Transmission and Break In Interference Tests



TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - NORM
2W 4W TEL - 2W
VF 20 - VF
MODE ONE MODE TWO - MODE ONE

Notes

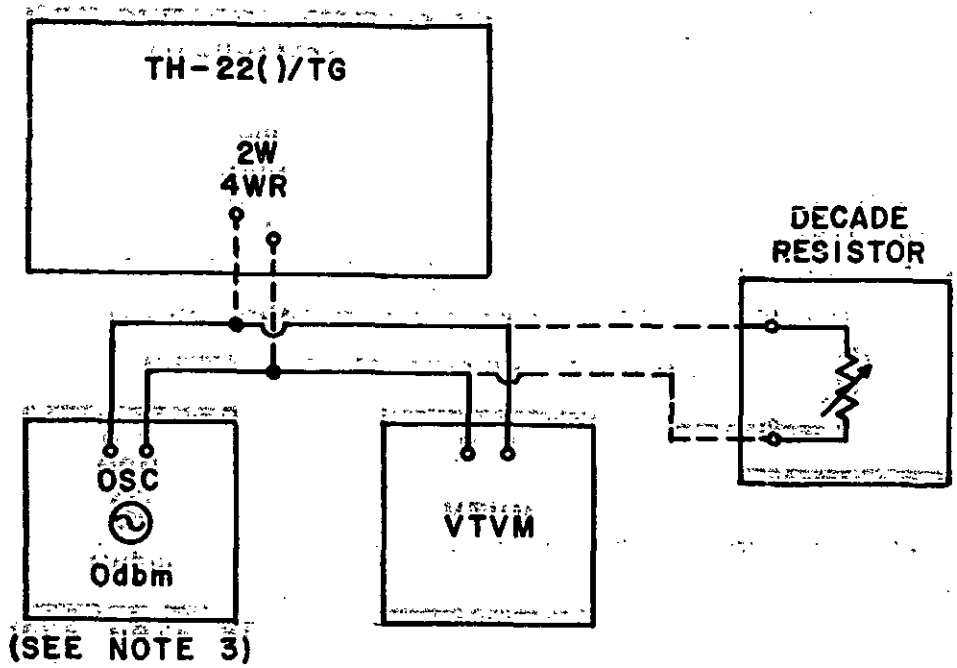
1. See Figure 1 for test equipment descriptions.

2. The oscilloscope, the frequency counter and the two telephones TA-312( )/U are not required for all tests. See applicable test paragraphs.

Figure 12

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TH-22( )/TG
Receive Circuit Impedance Test



TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - NORM
4W 2W TEL - Both (See Note 2)
VF 20 - VF
MODE ONE MODE TWO - MODE ONE

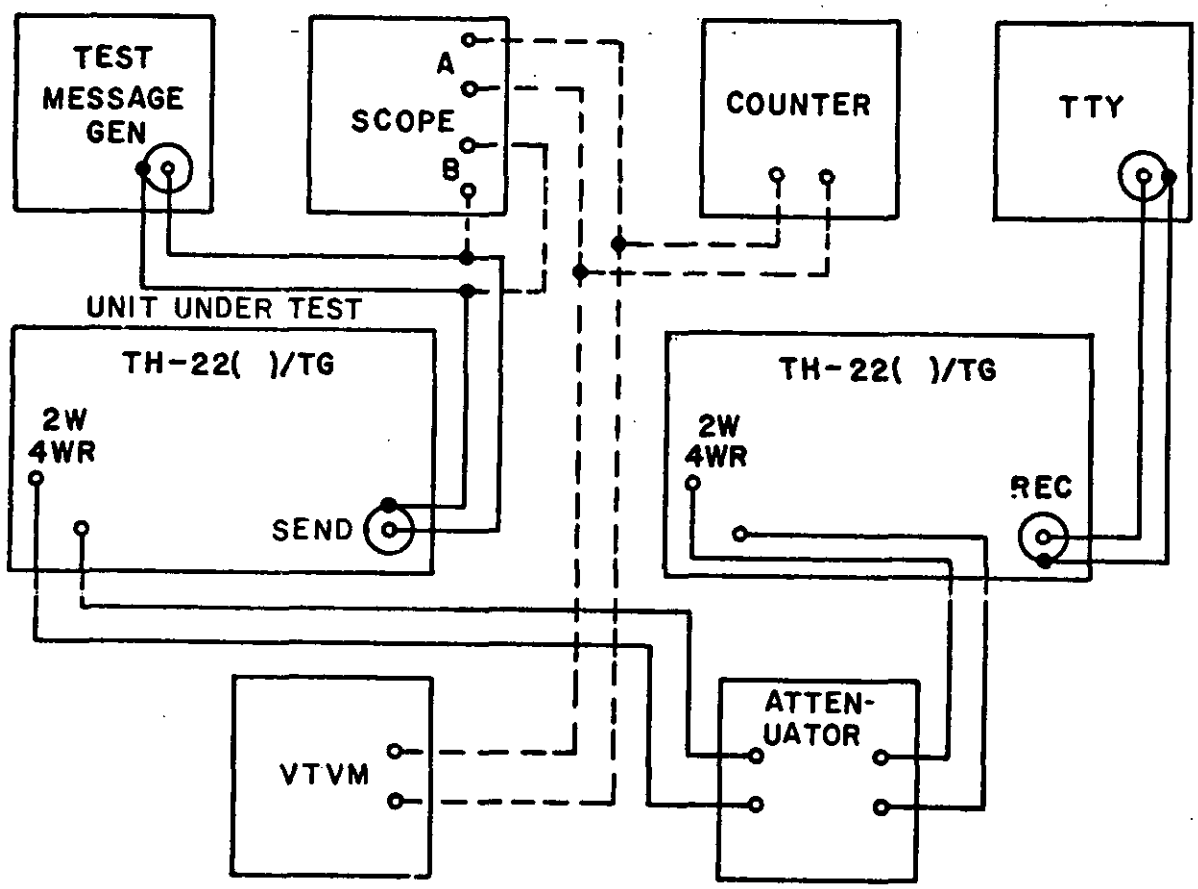
Notes

1. See Figure 1 for test equipment description.
2. Test to be conducted with switch in 2W position and then repeated with the switch in 4W position.
3. Test to be conducted at 1000 Hz and 2000 Hz.

Figure 13

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TH-22( )/TG
Break in Reception, Sensitivity, Time Delay and Reset Tests



TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - NORM
4W 2W TEL - 2W
VF 20 - Either
MODE ONE MODE TWO - MODE ONE

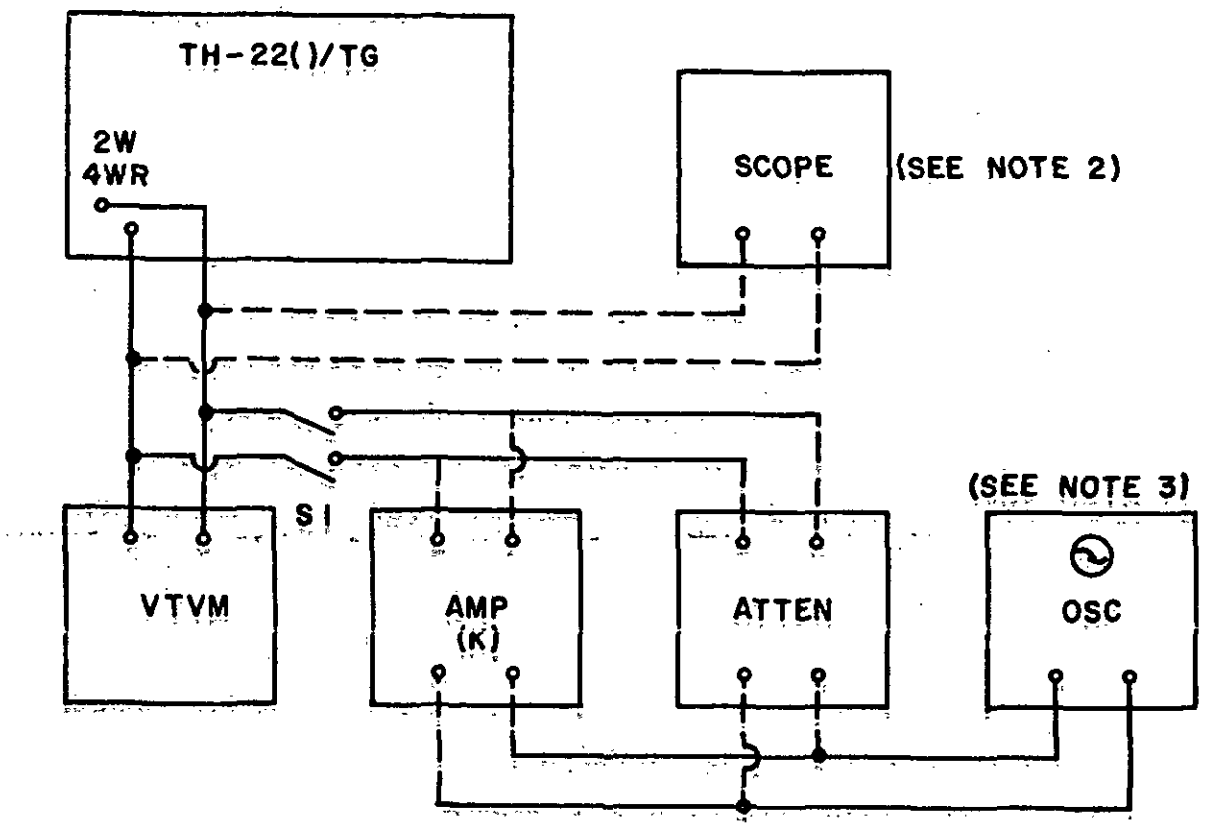
Notes

1. See Figure 1 for test equipment descriptions.

Figure 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:15Z
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TH-22( )/TG
20 Hz and VF Alarm Tests



TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC

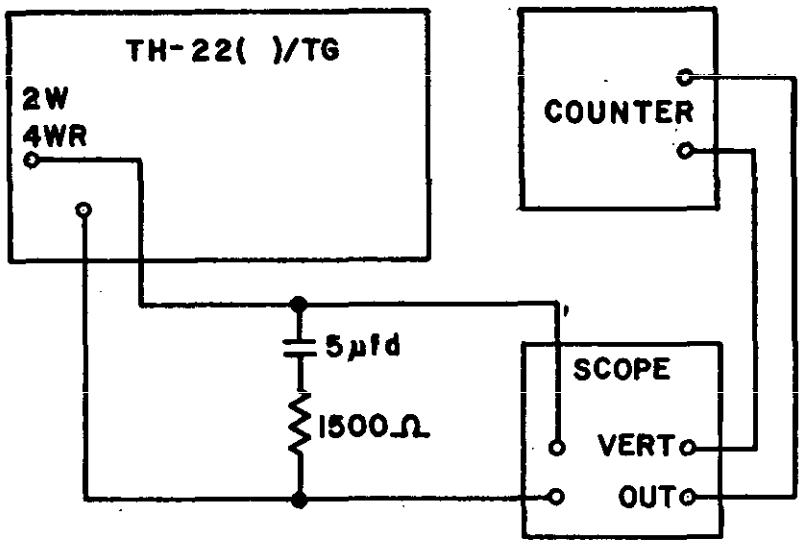
ON OFF - ON
NORM SEND REC - NORM
4W 2W TEL - See Note 4
VF 20 - See Note 5
MODE ONE MODE TWO - MODE ONE

Notes
1. See Figure 1 for test equipment descriptions.
2. Oscilloscope required for time delay measurement only.
3. Adjust oscillator to 1232.5Hz for VF alarm test and 20Hz for 20Hz alarm test.
4. Tests to be conducted at both 2W and 4W positions in accordance with applicable test paragraphs.
5. Adjust switch to VF for VF alarm test and to 20 for 20Hz alarm test.

Figure 15

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:15Z
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TH-22( )/TG
20 Hz Signal Output Test



TH-22( )/TG Switch Positions
115VAC 230VAC 26VDC - 115VAC
ON OFF - ON
NORM REC SEND - REC
4W 2W TEL - 2W
VF20 - 20
MODE ONE MODE TWO - MODE ONE

Notes

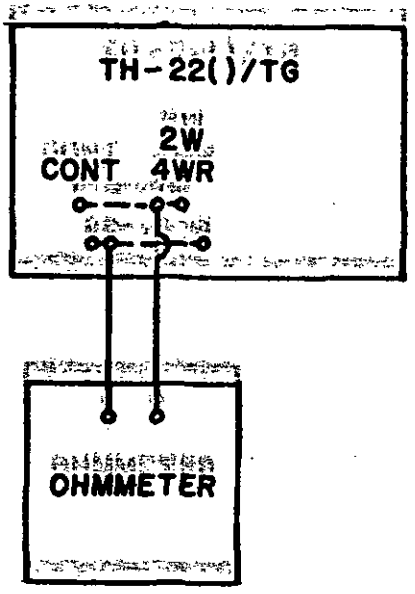
1. See Figure 1 for test equipment descriptions.

2. Resistor R60 shorted.

Figure 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:15Z
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TH-22( )/TG
Push to Talk Tests



TH-22( )/TG Switch Positions

115VAC 230VAC 26VDC - 115VAC
ON OFF - OFF (See Note 2)
NORM REC SEND - SEND
4W 2W TEL - See Note 3
VF 2O - Either
MODE ONE MODE TWO - Either

Notes

1. See Figure 1 for test equipment descriptions.

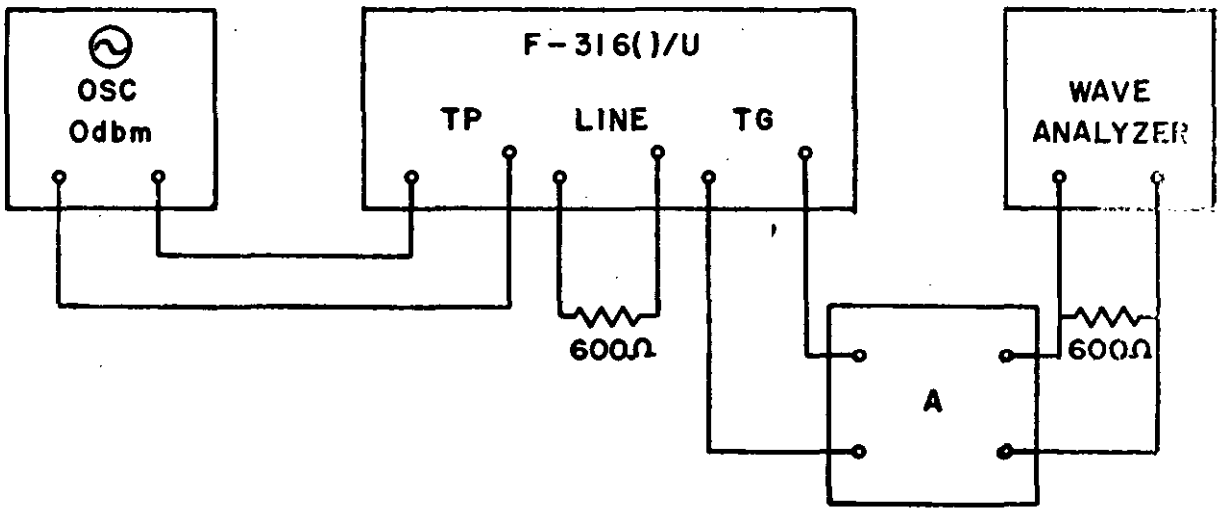
2. CAUTION: Insure that no power is applied to the TH-22( )/TG. A signal generated while resistance is being measured can damage the test equipment.

3. Set the switch to 2W when measuring resistance across the 2W 4WR terminals and to 4W when measuring resistance across the CONT terminals.

Figure 17

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:15Z
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F-316( )/U
Insertion Loss Test



Notes

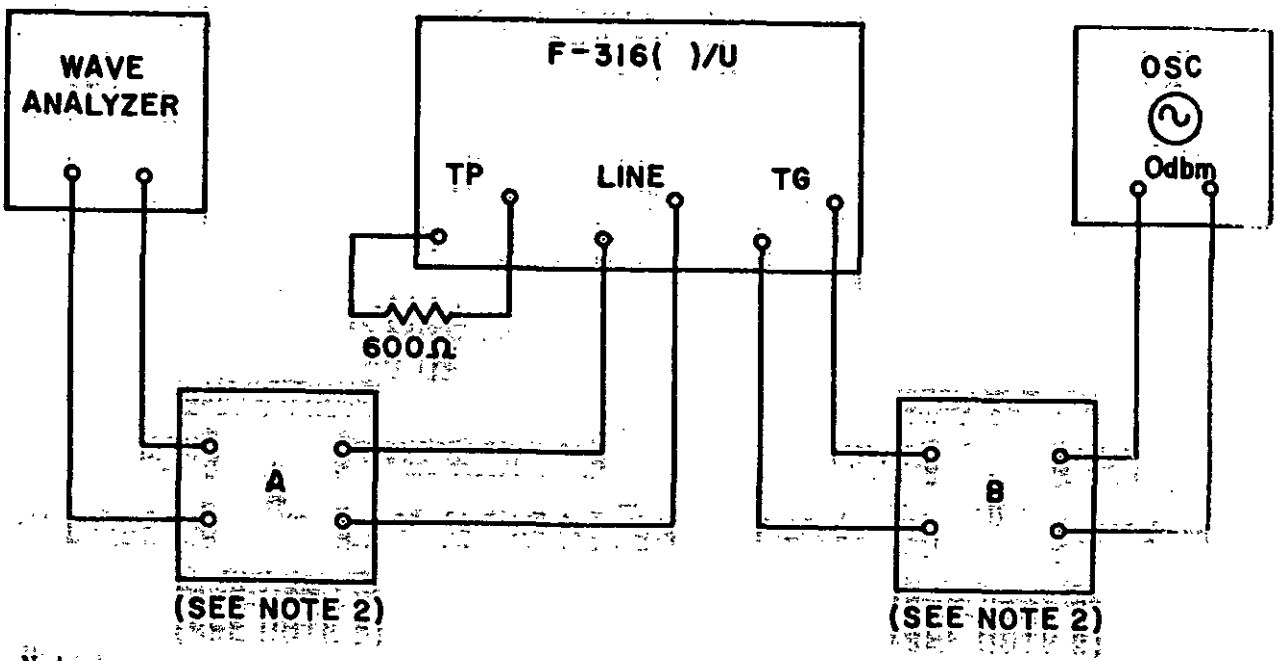
1. See Figure 1 for test equipment descriptions.
2. A is 600 ohm balanced isolation transformer.
3. Measure level at the TG binding posts at the following frequencies:

100 Hz	2000
500	2200
1000	2600
1200	2800
1400	3000
1600	3200
1800	3500

Figure 18

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:15Z
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F-316( )/U
Band Pass Test



Notes

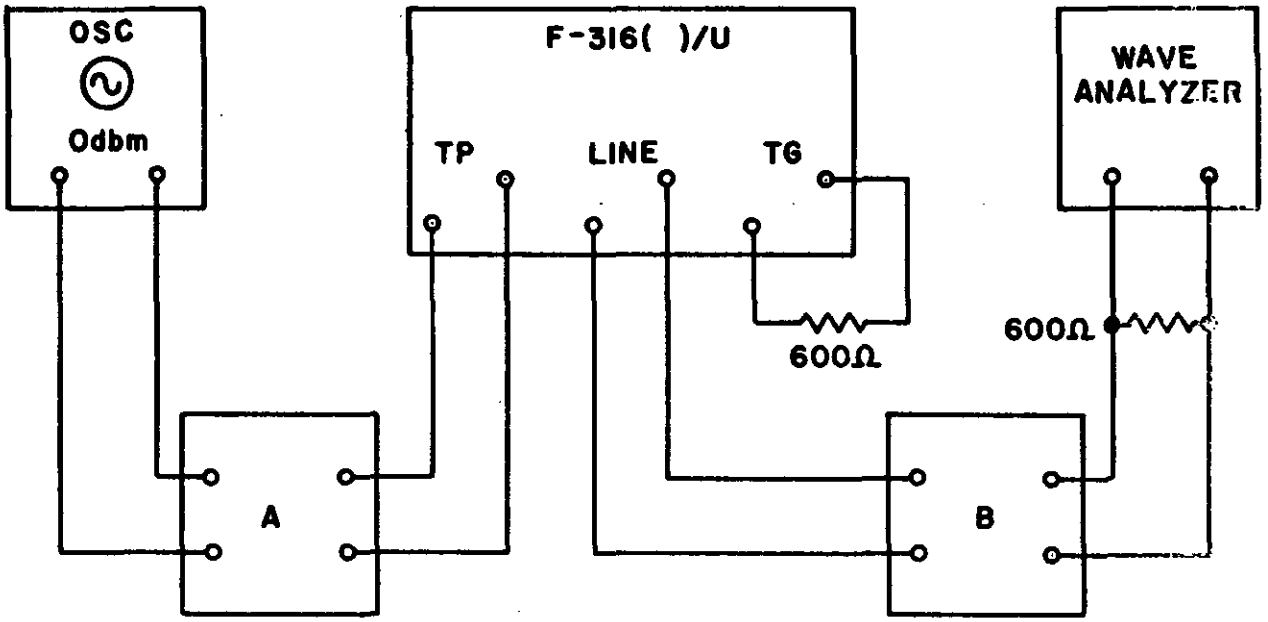
1. See Figure 1 for test equipment descriptions.
2. A & B are 600 ohm balanced isolation transformers.
3. Measure level at the LINE binding posts at the following frequencies:

Sweep 100 Hz - 1000 Hz
1180 Hz
1225 Hz
1232.5 Hz
1317.5 Hz
1325 Hz
1370 Hz
Sweep 1600 Hz - 3500 Hz

Figure 19

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:15Z
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F-316( )/U
Band Stop Test



Notes

1. See Figure 1 for test equipment descriptions.
2. A & B are 600 ohm balanced isolation transformers.
3. Measure level at the LINE binding posts at the following frequencies:

Sweep 1000HZ - 1000 Hz

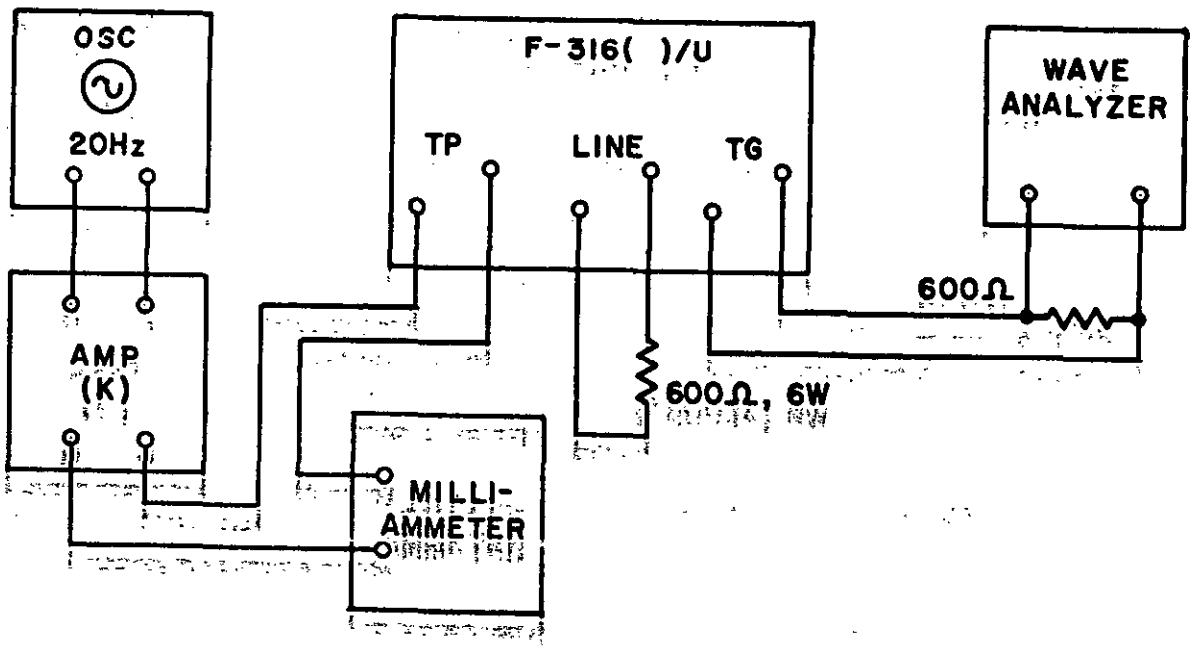
Sweep 1180HZ - 1370 Hz

Sweep 1600HZ - 3500 Hz

Figure 20

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:15Z
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F-316( )/U
Residual Effects Test



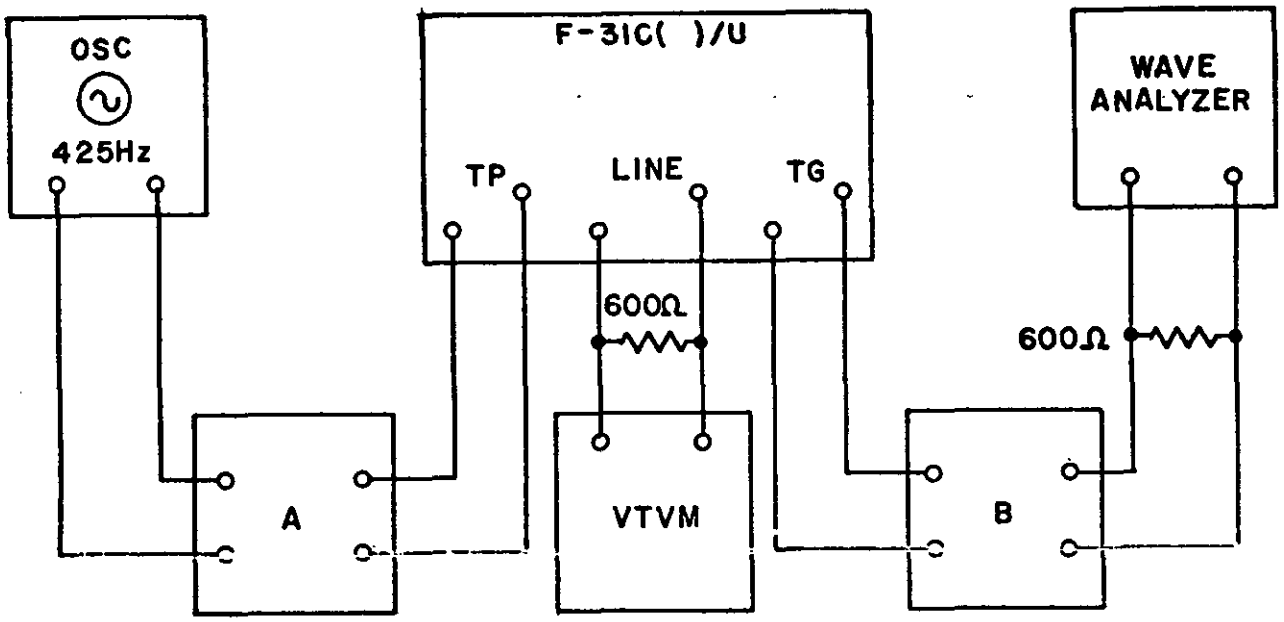
Notes

1. See Figure 1 for test equipment descriptions.
2. Adjust amplifier to 100 ma input to filter for one minute.

Figure 21

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:15Z
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F-316( )/U
Crosstalk Test



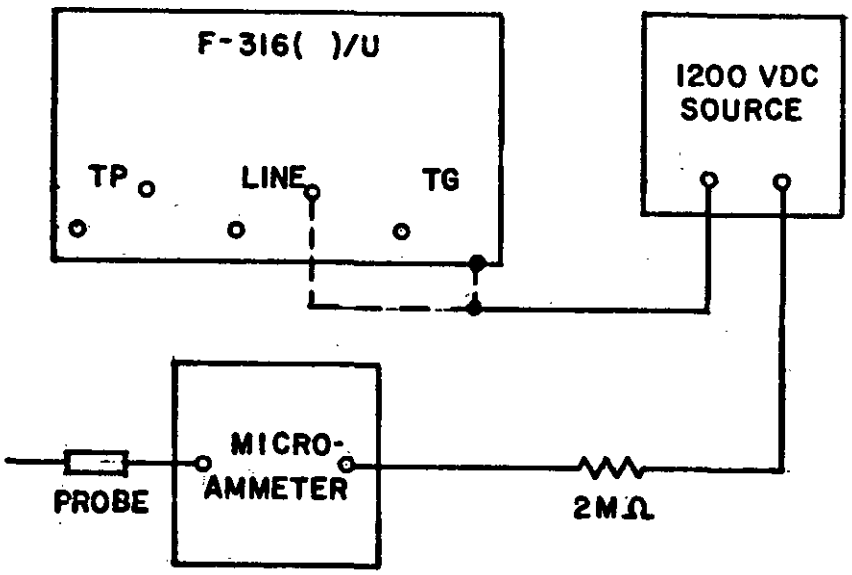
Notes

1. See Figure 1 for test equipment descriptions.
2. A and B are 600 ohm balanced isolation transformers.

Figure 22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:15Z
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F-316( )/U
Dielectric Strength Test



Notes

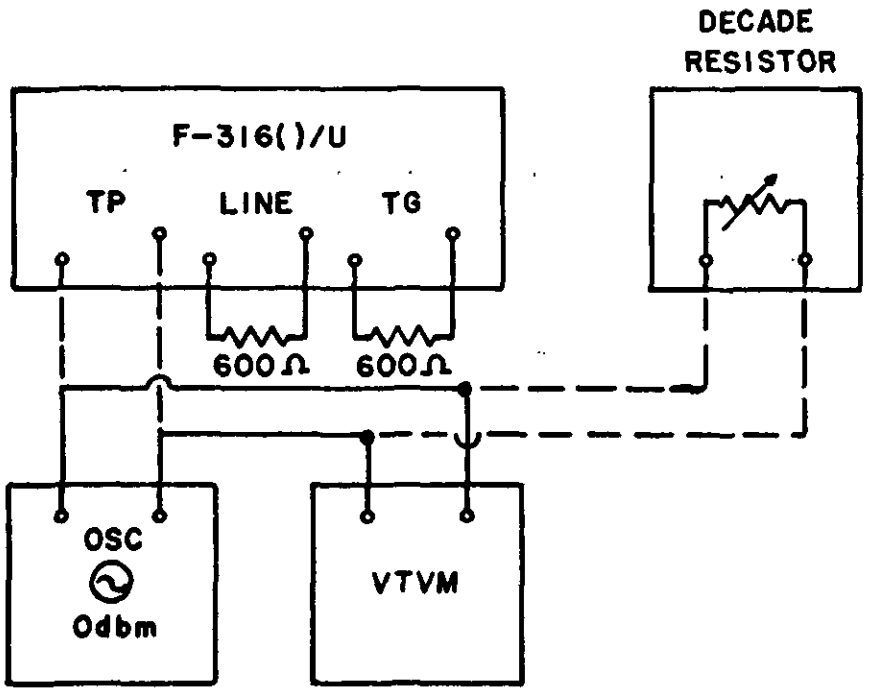
1. See Figure 1 for test equipment descriptions.

2. CAUTION. - Discharge capacity between each terminal and case into a 2 megohm load at completion of measurement.

Figure 23

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:15Z
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F-316( )/U
Impedance Test



Notes

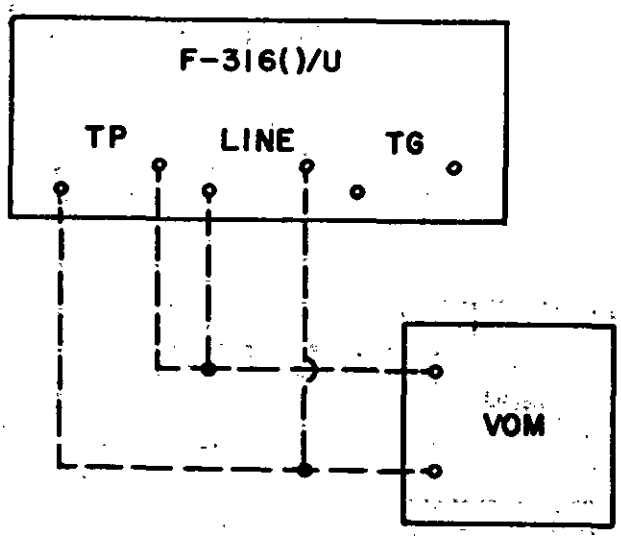
1. See Figure 1 for test equipment descriptions.

2. Repeat test applying signal to TG terminals with 600 ohm resistors connected across TP terminals and LINE terminals. Repeat test applying signal to LINE terminals with 600 ohm resistors connected across TP terminals and TG terminals.

Figure 24

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F-316( )/U
Balance Test



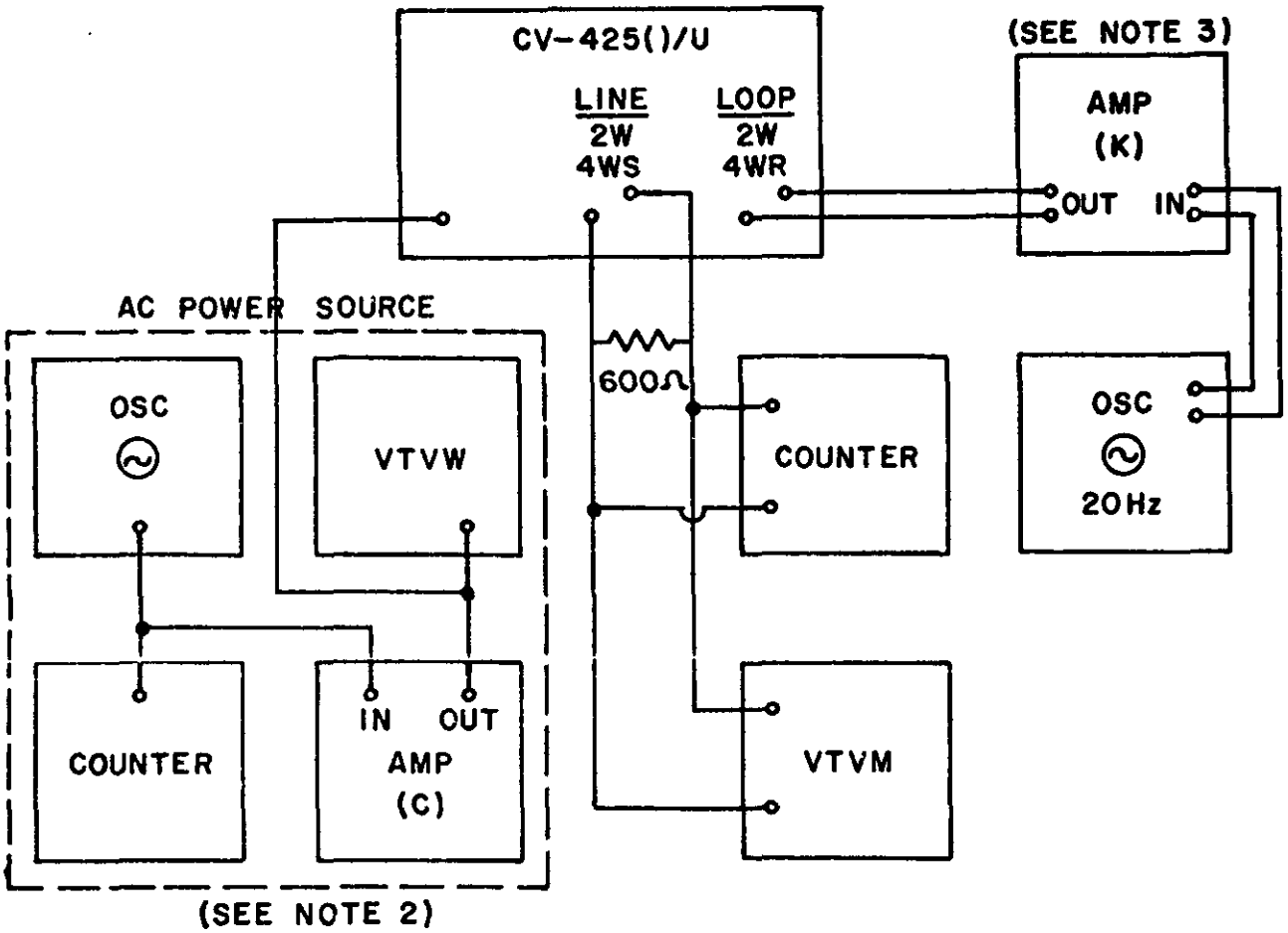
Notes

1. See Figure 1 for test equipment descriptions.

Figure 25

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CV-425( )/U
Power Source, VF Output, VF Level and 20Hz Sensitivity Tests
and Operational Inspection

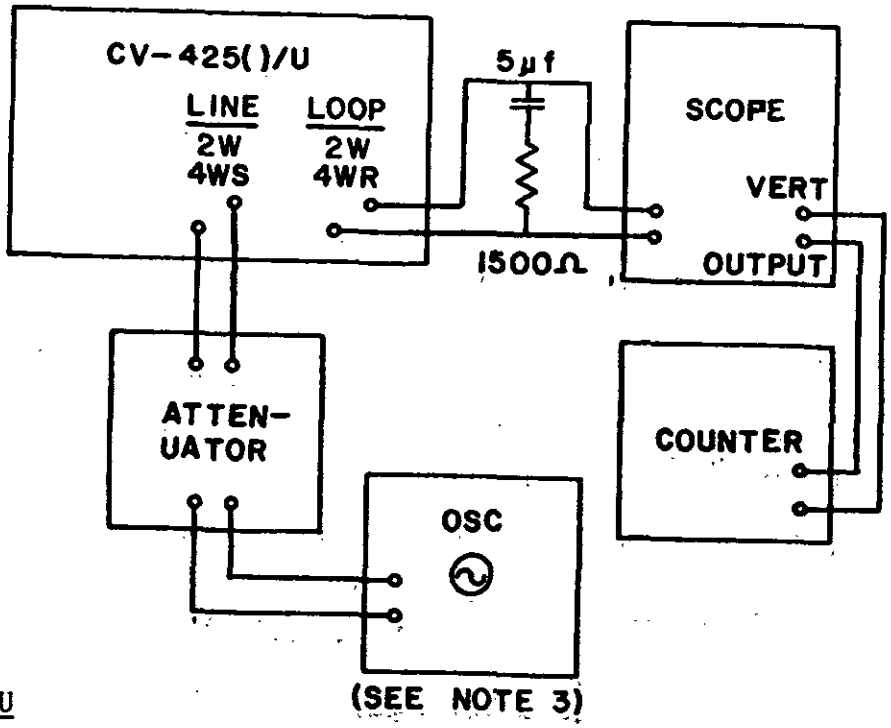


CV-425( )/U Switch Positions
POWER - ON
TP TG - Both (See Note 4)
2W 4W - 2W
SENSITIVITY - Either

Notes
1. See Figure 1 for test equipment descriptions.
2. Power source oscillator frequency and amplifier output level to be adjusted in accordance with power source test paragraph. Variable ac power source, not required for operational inspection.
3. 20Hz amplifier output level to be adjusted to 180 volts peak to peak for power source and VF output test and 50 volts peak to peak for 20Hz sensitivity test and operational inspection.
4. TP TG switch set to TG position to measure 1232.5Hz output frequency and level and to TP position to measure 1600Hz output frequency and level.

Figure 26

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CV-425( )/U
POWER - ON
TP TG - Both (See Note 3)
2W 4W - 2W
SENSITIVITY - Both (See Note 3)

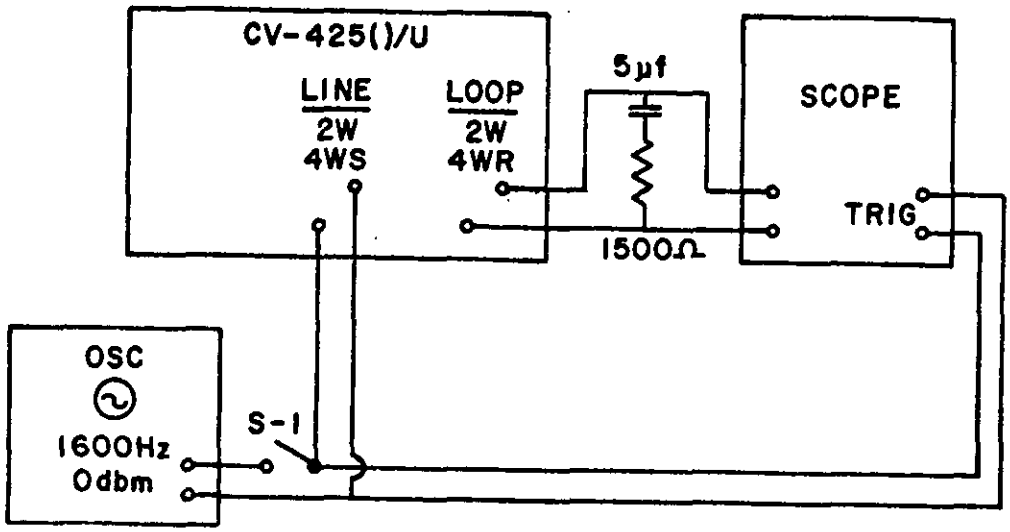
Notes

1. See Figure 1 for test equipment descriptions.
2. Resistor R6 shorted.
3. TP TG and SENSITIVITY Switch positions, oscillator frequency and attenuator level to be adjusted in accordance with 20Hz output and VF sensitivity test paragraph or operational inspection test paragraph.

Figure 27

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CV-425( )/U
VF Guard Circuit Test



CV-425( )/U Switch Positions
POWER - ON
TF TG - TP
2W 4W - 2W
SENSITIVITY - HIGH

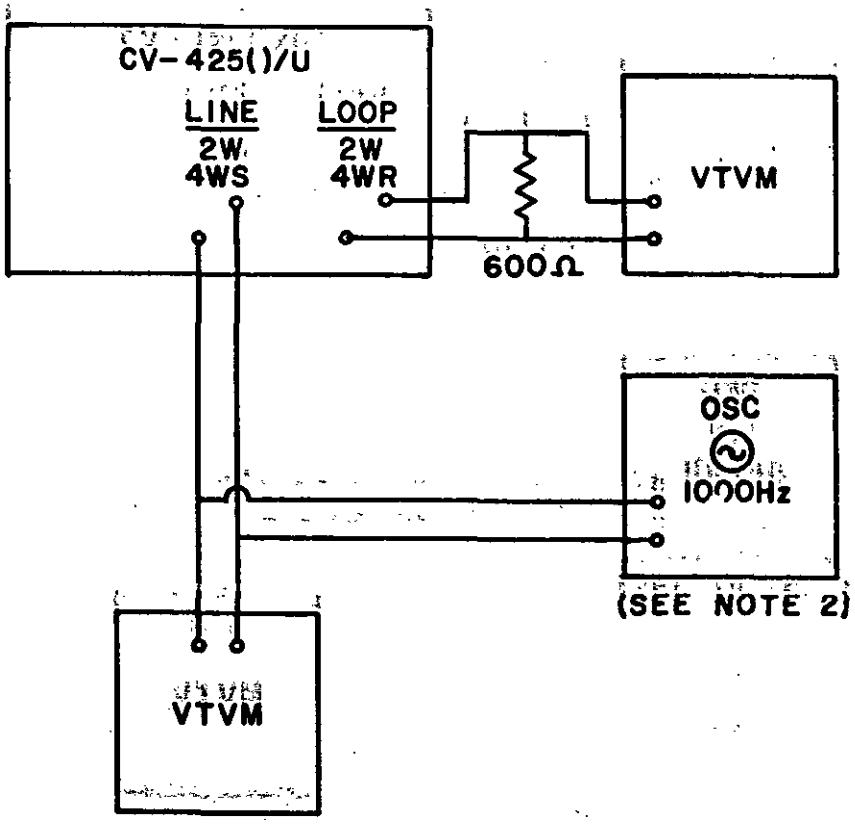
Notes

1. See Figure 1 for test equipment descriptions.

Figure 28

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CV-425( )/U
Signal Loss Test



CV-425( )/U Switch Positions
POWER - ON
TP TG - Either
2W 4W - 2W
SENSITIVITY - HIGH

Notes

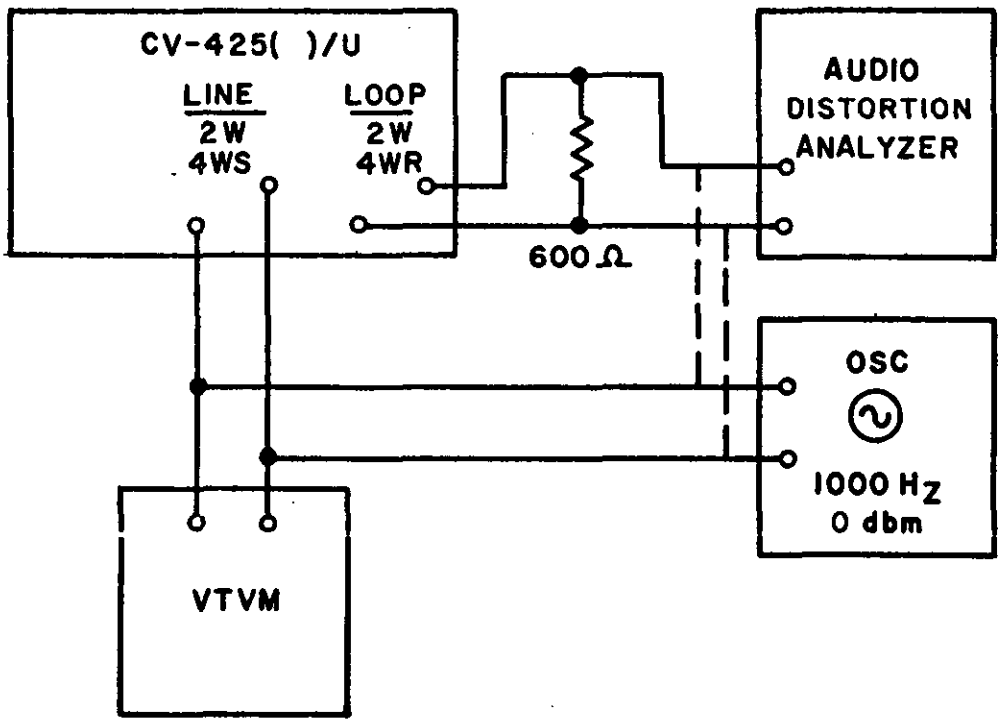
1. See Figure 1 for test equipment description.

2. Oscillator level to be adjusted to 0dbm and to -40 dbm in accordance with signal loss test paragraph.

Figure 29

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CV-425( )/U
Distortion Test



CV-425( )/U Switch Positions
POWER - ON
TP TG - TP
2W 4W - 2W
SENSITIVITY - HIGH

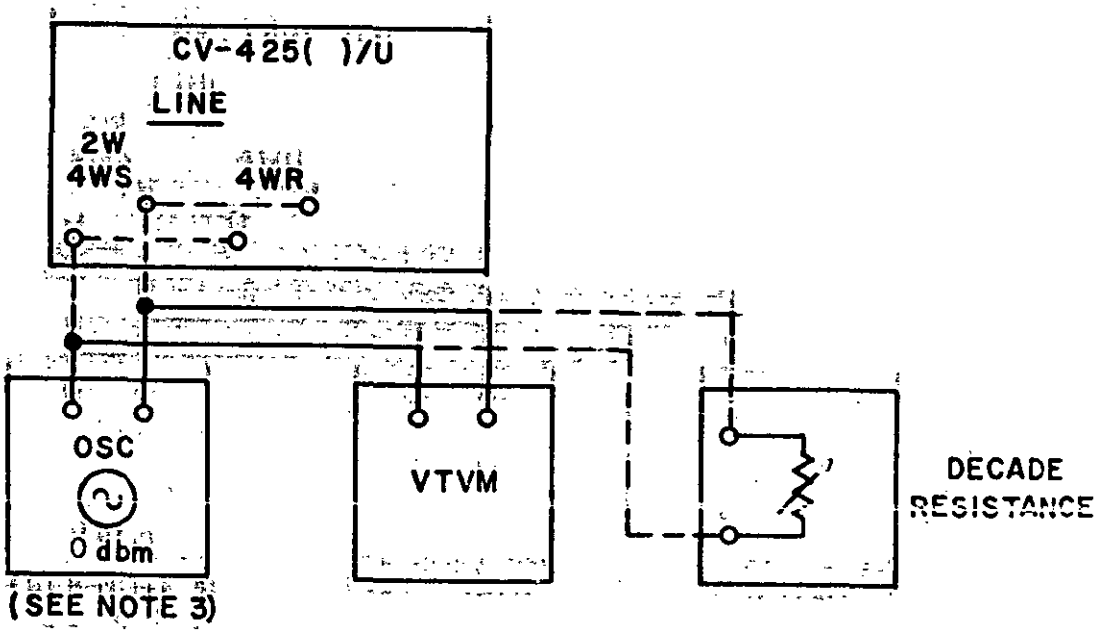
Notes

1. See Figure 1 for test equipment descriptions.

Figure 30

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CV-425( )/U
Line Input Impedance Test



CV-425( )/U Switch Positions
POWER - ON
TP TG - TP
2W 4W - Both (See Note 2)
SENSITIVITY - HIGH

Notes

1. See Figure 1 for test equipment descriptions.

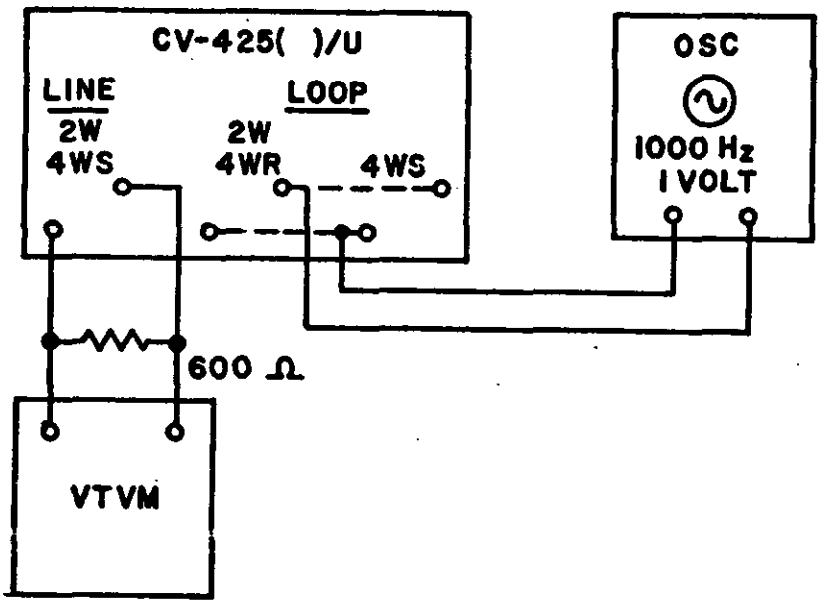
2. Test to be conducted with oscillator connected to 2W 4WS LINE binding posts with the 2W 4W switch in the 2W position. Test is then repeated with the oscillator connected to the 4WR LINE binding posts and the 2W 4W switch in the 4W position.

3. Test to be conducted with oscillator frequency adjusted to 200, 1000 and 3500 Hz.

Figure 31

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CV-425( )/U
Line Output Impedance Test



CV-425( )/U Switch Positions
POWER - ON
TP TG - TP
2W 4W - Both (See Note 2)
SENSITIVITY - HIGH

Notes

1. See Figure 1 for test equipment descriptions.

2. Test to be conducted with oscillator connected to 2W 4WR LOOP binding posts and 2W 4W switch in the 2W position. Test is then repeated with the oscillator connected to the 4WS LOOP binding post, and the 2W 4W switch in the 4W position.

Figure 32

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-T-55255A(EL) TERMINAL, TELEGRAPH-TELEPHONE AN/TCC-29( )
AND UNITS OF

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

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